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Competencies for teaching with and about artificial intelligence in the natural sciences — DiKoLAN AI

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ABSTRACT

The rapid advancement and widespread adoption of digital technologies have transformed the education sector. Among these developments, the emergence of generative artificial intelligence (AI) tools such as ChatGPT has had a considerable impact on teaching and learning practices. While the integration of AI into educational settings is becoming increasingly common, subject-specific analyses, especially in STEM education, are still lacking. This paper examines the specific challenges and potential of AI in the context of STEM education. It does so by exploring how AI has transformed scientific disciplines and how these changes impact teaching and learning. It highlights the necessity for educators to acquire specific competencies to effectively incorporate AI into their instructional practices. Building on existing frameworks such as DigCompEdu and the subject-specific DiKoLAN, the paper proposes an AI-focused framework: DiKoLAN AI. This framework aligns AI-related teacher competencies with instructional practice in science education. It also provides a structure for categorizing existing teacher training programs. The paper outlines the development of the DiKoLAN AI framework and its content consensus validation by a total of 64 experts through three iterative cycles. Its practical application is demonstrated through 20 case studies from different authors, which offer a practical approach for supporting teacher training and curriculum design in AI-integrated STEM education. The paper concludes with a discussion of opportunities, challenges and future research needs for teacher professionalization.

1. Introduction

Digital technologies have gained significant importance and interest in the education sector, particularly in the aftermath of the global pandemic caused by the coronavirus (SARS-CoV-2). This established the foundation for a long-term transformation of the educational system. Many of the tools and workflows developed during this period have been adopted on a permanent basis due to the positive experiences of users. However, a salient feature of digitalization is the accelerating rate of change it stimulates, which gives rise to profound societal transformations in domains such as the economy, medicine, and education. An illustrative example of this phenomenon can be found in the advent of ChatGPT, the first popular generative artificial intelligence (AI) software that gained widespread prominence: upon its launch, ChatGPT not only encountered an infrastructure that was well-suited to support digital applications and processes. It also encountered users who had become well-versed in the use and integration of digital tools into their daily lives. For this reason, among others, ChatGPT rapidly permeated numerous domains of society and presented novel challenges to the education system and other domains.

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The increasing and now widespread availability of AI applications has resulted in both learners and teachers utilizing AI for learning and teaching purposes. The advent of ChatGPT has further facilitated the integration of generative AI in educational settings, impacting various facets of teaching and learning, including text-based and graphics-based tasks and assessments. This phenomenon is already exerting a discernible influence on educational practices, strategies, and research trajectories across diverse disciplines and subjects [1].

However, it is evident that subject-specific or domain-specific analyses, for example in science education, with regard to the role and integration of AI in teaching, are also essential in order to derive the resulting opportunities, challenges and consequences for sustainable teaching [2]. On the one hand, subject-specific aspects emerge in relation to, as the integration of AI, as key component of contemporary scientific research, is becoming increasingly meaningful across various areas (cf. Nobel Prize in Chemistry and Physics 2024). On the other hand, adaptive learning environments are available for the respective field, resources to analyze and understand STEM learning with automated assessments in relation to learning progress and learning behavior [2,3]. The basis for high-quality STEM education is the consideration of complex social and pedagogical factors as well as the contexts and conditions in subject teaching and not just the use of AI technologies in education [2,4]. The disparities in the adoption of AI that are currently evident in the subjects taught in the area of science education are also reflected in a study on the impact of AI in teaching and learning science. For instance, there is a notable lack of studies in the field of chemistry compared to biology and physics [5]. Moreover, the goals associated with integrating AI, ranging from conceptual and epistemic to social dimensions of science learning, also vary considerably [6]. Therefore, it is evident that the competencies required of science teachers are extensive.

In order to realize the hoped-for benefits of AI use in practice, an examination of the role and competencies of teachers is necessary. It is therefore essential to identify teacher competencies that are relevant in the context of AI use. Markauskaite et al. [7] inquire into the competencies educators require to incorporate AI into their pedagogy and to prepare their students for an AI-driven world and future. Ng et al. [8] summarize with reference to Huang [9], that the lack of appropriate AI-related competencies among teachers has implications for the use of such tools, especially since teachers who are more competent in the use of AI-based technologies are more likely to adapt to the digital transformation, facilitating their teaching and administrative work while taking challenges into account. Furthermore, they posit that curricula and theoretical frameworks serve as pivotal supports for educators in identifying the requisite AI competencies that are instrumental in facilitating teaching practices. In light of these considerations, they are developing additions and adaptations for DigCompEdu from a perspective that prioritizes AI-related competencies. These frameworks also serve as a foundation for the categorization of AI-related teacher training [e.g., 10], particularly in terms of developing coherent competency profiles and designing curricula for future teacher education. This is of particular importance, as teachers are often able to recognize and articulate their own deficits and needs, and frequently demonstrate a willingness to engage in further professional development in this area [11]. Consequently, the development of training programs should be carefully structured to facilitate targeted teacher training and further education.

However, there is currently a lack of established frameworks or conceptual models in educational theory or educational research, and there is also a lack of a sufficiently robust body of data to address this challenge. Neither the meaning nor the implications of AI for subject-specific teaching have been adequately clarified. To determine which aspects of AI technologies are relevant for teaching, particularly for subject-based instruction and teacher education, it is essential, as a first step, to analyze how AI is transforming, or has already transformed, relevant domains.

Given the diverse subject-specific requirements and instructional structures, AI-related competencies and corresponding framework models for teachers should incorporate a certain degree of subject-specificity. In this regard, it is logical to extend the approach of Ng et al. [8] by adopting a more subject-oriented and pedagogical perspective. A similar approach is seen in the DiKoLAN framework [12], which refines general digitalization-related skills (as structured in DigCompEdu) for science teachers. Based on the TPACK model, DiKoLAN formulates teacher competencies with direct reference to specific teaching-related domains of action. Just as DigCompEdu has been adapted with a dedicated focus on AI, a comparable adaptation of DiKoLAN is also feasible. The resulting AI-related framework, DiKoLAN AI, can then be utilized to categorize existing AI-related training and further education programs for (prospective) teachers.

Unlike general competence frameworks such as DigCompEdu, which address digital competences in a broad and cross-disciplinary manner, DiKoLAN AI is explicitly developed for the subject-specific context of science education. It focuses on AI-related competencies and integrates three interdependent dimensions: didactic competencies for designing meaningful AI-supported learning processes, technological competencies for effectively using and understanding AI tools, and socio-cultural competencies for critically reflecting on AI's societal implications. This triad constitutes a key strength of the framework, as it enables a holistic approach to preparing science teachers for the pedagogical, technical, and ethical challenges of AI integration.

In addition, DiKoLAN AI is designed for flexible application across multiple contexts in science teacher education. It can be used as a guiding framework for curriculum development, as a tool for teacher self-assessment, and as a basis for structuring targeted professional development offerings, thereby supporting both educators and policymakers in systematically integrating AI into science education.

2. Research objectives and contribution

This paper aims to address the challenges and opportunities associated with the integration of AI into STEM education by identifying subject-specific teacher competencies and appropriate teacher training frameworks. The primary objective is to develop and present *DiKoLAN AI*, a framework that expands the existing DiKoLAN competence framework to include AI-related dimensions relevant to science education. To this end, the paper

1. analyses how AI is transforming the STEM disciplines themselves,
2. explores the implications of AI for teaching and learning in STEM subjects, and
3. evaluates existing teacher competence frameworks with a focus on AI integration.

Based on these analyses and on an three step-iterative expert validation, the paper

4. introduces the DiKoLAN AI framework.
5. demonstrates the practical applicability of DiKoLAN AI by providing a series of case studies. These case studies are intended to illustrate the relevance of the DiKoLAN AI framework for the design and analysis of teacher training programs.

3. Background

3.1. AI in natural science research and application

Specific aspects of AI integration into science education are associated with its implementation in contemporary scientific research. Given the integration of research methods within the science curriculum, the related knowledge and competencies are essential for science teachers. Therefore, the following examples of AI applications are presented in

Table 1. These examples draw from the fields of chemistry, physics, and biology, as well as from interdisciplinary fields such as physical chemistry and biochemistry. The examples highlight the extensive range of potential applications and the interdisciplinary character of AI technologies. Of particular note are the prediction of biological and chemical processes, the design of new materials and medicines, and the automation and optimization of laboratory processes, which are currently key areas of research and application in the natural sciences. The significant entry and advance of AI in specialized sciences should also be integrated into the corresponding subject teaching by contextualizing the learning content in alignment with further developments in the field of AI. Several context can be identified for science subjects that facilitate authentic and socially relevant learning. This is illustrated in Table 1 through selected examples from the natural sciences. The table also includes the connection to the competency areas described in the DiKoLAN AI framework (see Section 5.1).

AI is one of the ten most important emerging technologies in chemistry in 2023 [13]. The influence of AI has grown significantly across all disciplines and has the potential to fundamentally transform various industries. It can be expected that, similar to the AlphaFold software, several AI tools will be developed that will be of fundamental importance for the natural sciences.

Advancements in technology have led to continuous improvements in the field of AI, resulting in increasingly efficient, reliable, and fast AI technology. The use of new supercomputers and big data technologies has enabled the interpretation of large data sets using machine learning (ML) and neural networks, a process that is critical for numerous academic disciplines [14,15]. Without the application of various AI methods, analyzing this vast amount of data would require enormous expertise, significant financial expenditure, and considerable time [16].

As demonstrated in their review, Berber et al. [17] have shown that nearly all areas of scientific research in chemistry, physics, and biology utilize AI applications as part of their research. The authors also indicated in which area of teaching, according to the DiKoLAN AI framework, these applications from scientific research can be categorized (an exemplary overview of this can be found in Table 1). These examples highlight the wide range of potential applications and the interdisciplinary character of AI technologies. Of particular interest are the prediction of biological and chemical processes, the design of new materials and medicines, and the automation and optimization of laboratory processes, which are central to current research and application in the natural sciences. Several contexts can be identified for science subjects that facilitate authentic and socially relevant learning.

3.2. AI in education and teacher competencies

In recent years, a growing body of literature and frameworks have emerged that examine the disruptive role of AI in education [e.g., 45, 46]. It has been emphasized that key principles such as fairness, accountability, transparency, and ethics are crucial for the effective and equitable use of AI in education [e.g., 47].

Generally, this technology holds significant potential for improving teaching and learning processes [e.g., 46,48]. Specifically, these applications are expected to assist in addressing significant challenges within educational systems, including diversity, inclusion, and teacher shortages [e.g., 49]. These applications have the potential to provide students with the necessary support and customization in their learning processes, thereby alleviating teachers of some of their responsibility [e.g., 46,50,51].

However, it is crucial to recognize that these opportunities come with risks that should not be overlooked. Key concerns include data protection and security issues, as well as the possibility of certain groups being discriminated against due to inaccurate training data. In educational settings, where students represent a particularly vulnerable population, there is an urgent need for AI systems that are trustworthy and capable of making fair decisions. Existing systems can be optimized, and Learning Management Systems (LMS) can be employed to innovate beyond the simple provision of online learning content. For instance, they can facilitate the formation of learning groups that are interactive and adaptable to different situations based on skills or other relevant characteristics [52]. Innovative massive open online courses (MOOCs) can offer continuous AI support that adapts to individual learning processes without interruptions for assessment. This adaptability is possible globally, transcending language barriers and financial limitations [53], which aligns with the fourth goal of the Sustainable Development Goals [54].

AI technologies are expected to play a vital role in educational data mining [55] and learning analytics [52], providing data-driven insights that can inform evidence-based educational practices. This can lead to more effective and efficient education methods, enabling the adaptation of teaching strategies and learning materials based on concrete data to maximize student success.

Despite AI's widespread use and obvious benefits in education, the teaching of AI-specific knowledge and the development of corresponding skills are not yet adequately addressed or integrated into subject-specific teacher training. The necessity of AI skills extend beyond a mere understanding of the technology; today, teachers must be able to use AI in a pedagogically and didactically meaningful and responsible manner within the classroom. This encompasses not only a comprehension of the technology itself but also an awareness of ethical considerations, data protection, and potential bias issues that are associated with AI systems [47]. In this regard, the European Commission [56] emphasizes the pivotal role teachers and school leaders play in implementing and utilizing AI systems, particularly in recognizing the benefits and addressing the challenges. The requirements associated with the use of AI in the classroom must be mastered by every teacher, regardless of subject, as there are areas of action that are rather subject-neutral. Conversely, there exist more subject-specific domains of activity in which the applications of AI can facilitate the learning process, such as in the fields of biology, chemistry, and physics. At the same time, however, teachers also represent a specific subject and, therefore, have the task of conveying subject-specific methodologies so that students gain an adequate understanding of the respective domain in broad terms. Consequently, all teachers are confronted with the task of systematically integrating teaching and learning with and about AI into their subject lessons. To fulfill their educational mandate, teachers must possess the relevant didactic, pedagogical, and technological skills and abilities regarding AI. However, the reality of teacher professionalization lags far behind these requirements. The integration of AI into the curriculum in a structured manner, and thus the cumulative promotion of AI skills among pre-service and in-service teachers, is not yet possible. There is a need for subject-specific approaches to the integration of digital technologies and skills for teaching subjects in general and AI in particular.

Given the rapidly evolving nature of AI technology and the proliferation of AI applications, it seems essential to ensure that teacher training programs adapt in a timely manner to capitalize on emerging opportunities and ensure a lasting impact on school practices. In addition to developing skills, this could reduce reservations and fears about this new technology among (prospective) teachers and increase the acceptance of AI as a teaching and learning tool as a prerequisite for its widespread use in the classroom [57]. The foundation for this cumulative skills development is the establishment of structured reference frameworks. Within these frameworks, the AI-related skills that (pre-service) teachers must acquire are systematically defined in subject-specific terms.

There are still no clear concepts that account for teacher tasks and competencies and apply to different subjects. This makes it increasingly difficult to assess future challenges in education accurately. From a didactic perspective, clear goals and requirements for using AI in schools should be defined at an early stage. This can prevent AI technologies from being introduced in an uncontrolled manner and later requiring correction or analysis great expense.

Table 1
Overview of AI applications in scientific research, categorized by application area and field of expertise, grouped into *Image & Data Analysis*, *Prediction*, and *Synthesis & Lab Work*.

	Application area	Function	Fields of expertise	DiKoLAN AI Association	Source
Image & Data Analysis	Histological or X-ray image processing	AI can be used to classify microscopic images or X-ray images, for example, to recognize lung cancer in a histological section or bacterial colonies on a culture plate.	Cancer diagnostics, diagnostics of bacterial colonies	AI.DAQI	[18–21]
	Literature database analysis	AI can predict future research priorities by searching and analyzing literature.	General	AI.ISE, AI.DAP, AI.SIM	[22]
		To process data for categorizations and predictions, AI systems can use the following data, for example:			
		Experimental procedures	General	AI.ISE, AI.DAP, AI.AFA	[23]
		Molecular and protein properties/structure	Pharmacology, drug development, protein design, ligand research	AI.SIM, AI.DAP, AI.ISE	[24–31]
		Synthesis conditions	Synthesis in Chemistry	AI.SIM, AI.DAP	[32]
Prediction	Predicting structures	Known protein structures are extracted from protein databases and used as training data for AI.	Pharmacology, drug development	AI.SIM, AI.DAP, AI.ISE	[24]
		Based on amino acid sequence comparisons and known structural behavior (extracted from protein databases) structures of different proteins can be predicted.	Pharmacology, drug development, protein design	AI.SIM, AI.DAP, AI.ISE	[24–30]
		Protein structures and binding affinities can be used to predict the binding affinity of different ligands. This can also be helpful in the search for chemical probes.	Pharmacology, drug development, ligand discovery	AI.SIM, AI.DAP, AI.ISE	[25,31]
	Predicting material properties	AI can significantly accelerate the development of new substances and materials.			
		Predicting energetic properties of nanomaterials (such as the formation and Fermi energies) using AI enables statements about their structural properties.	Nanoscience, material development	AI.SIM, AI.DAP	[33]
		Based on a metal's processing conditions, AI can predict its microstructures and, in a further step, help achieve its target structures.	Metallurgy, material development	AI.SIM, AI.DAP	[34]
	Predicting interference	AI determines the arrival time of coronal mass ejections based on various observational data to predict the occurrence of possible communication system disruptions.	Astrophysics	AI.SIM, AI.DAP	[35]
	Modeling particle collisions	Modeling of proton interactions in collisions using artificial neural networks; trained with experimental data from proton collisions, AI systems can make very precise predictions about the results of collisions with changed parameters and evaluate huge data sets from real experiments.	High energy physics	AI.SIM, AI.DAP	[36]
Synthesis & Lab Work	Planning synthesis	Planning of synthesis routes using retrosynthesis; AI calculates which reaction steps are best suited to synthesize a desired product based on the reactions of structurally similar substances.	Synthesis in Chemistry, Drug Design	AI.SIM, AI.DAP	[37–40]
	Creating substance databases and documenting laboratory activities	ML can be used to create a reaction database automatically. AI models can then use the information from such databases to predict the possible reactions in a heterogeneous catalysis system.	Catalysis in Chemistry	AI.DOC, AI.SIM, AI.DAP	[41]
		Virtual, AI-based laboratory voice assistants take over previously manual work steps: oral work instructions – such as the documentation of measured values and observations – are understood and implemented with high accuracy.	Synthesis in Chemistry	AI.DOC, AI.AFA, AI.DAP	[42,43]
	Performing laboratory activities	Autonomous laboratories such as 'AlphaFlow' and 'Coscientist' use AI-based functions to supplement the skills of human researchers; assistance is provided, for example, in optimizing reactions by carrying out literature research, setting laboratory equipment and reaction conditions, and programming autonomously.	Synthesis in Chemistry	AI.ISE, AI.AFA, AI.DAP	[23,44]

3.3. Existing frameworks and limitations of teachers' digital competencies frameworks

In recent decades, various frameworks have been developed to professionalize the digital skills of teachers. These frameworks continue to form the basis for the use of AI technology: Established models such as the PCK model (Pedagogical Content Knowledge; Shulman [58]), persist in their utility for the professional development of teachers and the targeted development of their teaching skills. The fundamental tasks and challenges inherent to the profession will persist into future, as will the didactic knowledge that supports teachers navigate the digital transformation. One of the best-known extensions of the PCK model in the domain of digital technological skills is the TPACK model [59]. This model describes how teachers combine technical knowledge (TK), pedagogical knowledge (PK), and subject knowledge (CK) to efficaciously conceptualize technology-enhanced lessons. The TPACK model demonstrates that the effective integration of digital technologies in the classroom environment necessitates not only technical expertise but also a profound comprehension of didactic methodologies and subject-specific content. The model has frequently been utilized to structure educational offerings and to systematize teachers, as the relevant dimensions of professional knowledge are defined and their interactions are well described. While the TPACK model offers a comprehensive framework for understanding teachers' knowledge domains, it remains ambiguous and vague regarding the specific competencies required by teachers in distinct subjects, such as biology or physics.

In the EU, the DigCompEdu competence framework [60] was subsequently implemented as a framework for describing and promoting digital competencies. The objective of this framework is to assist educators in preparing for the challenges posed by digital education and to provide them with guidance on the implementation of digital technologies in the classroom. The framework is comprised of six distinct areas of competence: Professional Engagement, Digital Resources, Teaching and Learning, Assessment and Feedback, Facilitating Learning, and Promoting Learners' Digital Competencies. These areas cover the professional utilization of digital technologies and their didactic application to facilitate learning processes. DigCompEdu also emphasizes that digital education extends beyond the mere utilization of technology, aiming to empower educators to foster learners' acquisition of digital skills, encompassing both general and subject-specific areas. The framework distinguishes between six competence levels, ranging from beginners to innovators, to enable individualized development. The primary function of DigCompEdu is to serve as a foundational framework for the development of national further training programs for teachers and has influenced educational strategies in many European nations. Overall, the competence framework guides to shape the digital transformation in education in a targeted and sustainable manner. Teachers' existing pedagogical and didactic skills must be supplemented with subject-specific digital skills, as set out in the DigCompEdu competence framework [60]. In accordance with DigCompEdu, this necessitates the differentiation and formulation of subject-specific competencies for teaching and learning, the utilization of digital resources, reflection and evaluation, and learner orientation. These competencies are essential for the digitally competent design of the teaching fields of action to be used in subject lessons.

3.4. The DiKoLAN framework

One such framework, which was originally developed and used in the DACH region (Germany, Austria and Switzerland), is the DiKoLAN framework (Kotzebue et al. [12], supplemented by the "Assessment, Feedback and Adaptivity" section by Meier et al. [61]). DiKoLAN, an acronym for "Digital Competencies for Teaching in the Natural Sciences", is a specially crafted competence framework designed to facilitate the digital education of teachers in the natural sciences. The objective of DiKoLAN is to equip science teachers with the competencies to utilize digital technologies effectively and subject-specifically in science lessons. To this end, the DiKoLAN framework defines eight central competence areas: "Assessment, Feedback and Adaptivity", "Documentation", "Presentation", "Communication/Collaboration", "Information Search and Evaluation", "Data Acquisition", "Data Processing", and "Simulation and Modeling". These areas are particularly relevant for science teaching (see Fig. 1). In DiKoLAN, for each of these eight competence areas, competencies in the knowledge domains of the TPACK model (TPACK, TPK, TCK, and TK) associated with TK are selectively considered for further structuring, following the taxonomy of name, describe, and apply. The result is a framework that contains very specifically defined competencies based on DigComEdu and TPACK. The appropriate combination enables pre-service teachers to act to promote learning in the pedagogical and didactic competence areas. For instance, in the context of digital assessment, as described in DigCompEdu (apart from the general principles of assessment and feedback), the identification and implementation of digital tasks, tools, communication channels, and forms, as well as data analyses, must be identified and implemented as a prerequisite for digital documentation and presentation for the transmission of (digital) feedback. Depending on the subject and the teaching processes, the technologies used (and their differences in the subjects) result in areas of competence that differ in terms of their degree of subject-specificity. Consequently, a comprehensive examination of the integration of these technologies within specific subjects becomes imperative.

A notable benefit of this approach is that it enables professionals from educational sciences, subject-specific fields, and subject-specific didactics to collaboratively identify competencies that are relevant to their shared curricular goals in teacher training. This structured framework assists in defining these competencies and can also be employed to clarify responsibilities within a conceptual competence grid.

3.5. Identifying gaps: what DigCompEdu, TPACK, and DiKoLAN miss in addressing AI in education

While existing frameworks such as DigCompEdu and TPACK provide valuable general structures for digital and pedagogical competences, they lack the necessary specificity to adequately address the complex and subject-related challenges posed by AI in science education. For example, DigCompEdu outlines digital competencies broadly but does not adequately address the disruptive and interdisciplinary nature of AI technologies. Similarly, TPACK fails to incorporate the socio-cultural and ethical dimensions that are increasingly important in AI-mediated teaching and learning contexts. The DiKoLAN framework sharpens these aspects for science education but was not originally designed to cover the unique demands of AI. Therefore, DiKoLAN AI extends and differentiates DiKoLAN by systematically identifying, categorizing, and operationalizing AI-related competences across teaching domains—ensuring that science educators are equipped not only with technical and didactic knowledge, but also with a critical understanding of AI's transformative impact on scientific practice and classroom instruction.

3.6. Frameworks that include teachers AI competencies

The integration of AI into education imposes new demands on the technological, didactic, and subject-specific skills of teachers. While the TPACK (Technological Pedagogical Content Knowledge) model persists as a foundational framework for teachers' professional knowledge, it is becoming increasingly evident that it is inadequate to encompass the numerous dimensions of AI in education [62]. Consequently, there are various approaches to extend and adapt TPACK specifically for the use of AI.

One approach that has been proposed is the AI-TPACK model, as stated by Ning et al. [63]. This model is based on the classic TPACK but integrates an "additional" knowledge dimension: AI knowledge (AI-TK). This is considered a specification or subset of technological knowledge (TK). AI-TK comprises specific knowledge about AI technologies, how they work, and their potential and limitations in an educational context. In a similar vein, Karataş and Ataç [64] confirm these assertions, emphasizing the necessity for teachers to possess not only fundamental digital competencies but also to cultivate a profound comprehension of the manner in which AI-based instruments can be employed to support learning processes.

Another concept that extends the TPACK model is Intelligent TPACK by Celik [65]. This framework incorporates AI-specific aspects into the traditional TPACK model, including the pedagogical possibilities of AI-supported tools such as automated feedback and personalized learning. The integration of AI in educational settings presents a multitude of opportunities to alleviate teacher's workload and provide customized support to learners. For instance, AI-supported systems can analyze students' learning progress in real-time and suggest adaptive learning paths. However, using such technologies also presents new challenges. Celik [65] emphasizes the necessity of addressing ethical concerns, in addition to the technical and didactic dimensions to ensure the responsible integration of AI in educational settings. These concerns encompass data protection, algorithmic bias, and the extent to which AI influences the autonomy of teachers and learners. These new models demonstrate that while TPACK remains a valuable basis for describing teachers' professional knowledge, it needs further development for the specific domain of AI application. Both models also indicate that there is no need to develop an entirely new model, but rather that an AI-relevant specification must be developed, with additional socio-cultural aspects also needing to be considered. From this, we conclude that a subject-specific version of AI competencies of science teachers does not require a new model, but rather an extension of existing frameworks or a specification of general digital competencies to artificial intelligence technology. The subject-specific variant of DiKoLAN, which we describe below, is to be understood in the same way as the TPACK extensions presented for taking AI competencies into account.

Fig. 1. DiKoLAN Framework according to Kotzebue et al. [12] supplemented by the "Assessment, Feedback and Adaptivity" section by Meier et al. [61].

4. Development of the DiKoLAN AI framework

4.1. Principles and development of the DiKoLAN AI framework

The DiKoLAN framework [12,61] is based on fundamental teaching fields of action rather than on specific or individual technologies. The principles relevant to data collection or data processing are not reliant on the technologies employed. However, technologies that are fundamentally different, or those with broad, unforeseen, and potentially disruptive impacts, require more in-depth consideration of the potential advantages and challenges in how teachers design and implement their lessons. To date, the DiKoLAN framework has not been expanded to address AI as a specific technology [61]. At the time of the DiKoLAN framework's publication in 2020, AI had not yet become a prominent element in school practice. Application areas such as learning analytics and adaptive learning environments, as well as the areas of assessment, feedback and adaptivity, were not yet significant fields for teacher training due to the limited or non-existent availability of systems in schools. As described in Section 3.3, the TPACK model underlying the DiKoLAN framework was extended to place a specific focus on AI. In the following, we describe this step of adaptation by illustrating how AI-related competencies can be differentiated and operationalized in concrete terms for science teachers, complementing the competencies already outlined in DiKoLAN with a specific emphasis on AI (DiKoLAN AI; see Fig. 2).

Given the significant explicit and implicit transformation processes that accompany advancements in AI, it is necessary to consider how the AI-related competencies of (pre-service and in-service) science teachers can be integrated into the classroom. Two distinct options exist for incorporating AI technology into the DiKoLAN framework:

- Option 1: Expansion of the DiKoLAN orientation framework to include an additional "Artificial Intelligence" competence area.
- Option 2: Integration of AI-related competencies into the individual DiKoLAN competence areas.

However, both options present challenges. "Artificial Intelligence" as a reference topic does not, in itself, constitute a teaching activity; it merely signifies a technology that can be applied in various fields of action. Therefore, it would be more logical to document relevant AI-related competencies as concrete, operationalized individual competencies within the DiKoLAN competence areas instead of as described in option 1. However, this second option also involves difficulties in terms of implementation. Unlike previous technologies, AI is a disruptive technology that has the potential to affect all areas of teaching, technological, societal, sociocultural, pedagogical, scientific, and didactic aspects of teaching. In this regard, AI-related competencies are also pervasive in relation to DiKoLAN competencies. AI has the potential to play an integral role in many existing competencies and to give rise to entirely new ones. From this perspective on AI, the competencies of the framework as a whole would therefore have to be revised and supplemented with new ones. This structural approach would considerably impede the targeted development of AI-related competencies in the professionalization of teachers. To address specific technologies, it would be appropriate to consider at the activities of teaching in the DiKoLAN framework from a specific perspectives – in this case AI – and to develop them for teaching and the professionalization of teachers with a focus on AI (as option 3).

Furthermore, to adequately address the various dimensions and extensive impacts of artificial intelligence within society, additional modifications to the foundational framework are necessary.

4.2. Artificial intelligence requires more than technical knowledge

In the early stages of digitalization, the primary focus was on the technological aspects and their implications for information, communication, and educational processes. However, with the widespread availability of data networks and mobile technologies, it has become clear that digital transformation cannot be regarded as purely a technological advancement. Instead, a more nuanced examination of the sociocultural and media dynamics of digitality is necessary to understand the challenges and opportunities for teaching and learning.

Recognizing the increasing importance of social and cultural factors in digitality, Thyssen et al. [66] have broadened the existing framework of technological knowledge (TK), pedagogical knowledge (PK), and

Fig. 2. DiKoLAN AI Framework on the surface structure. For better differentiation from DiKoLAN [12] and the specification on artificial intelligence, the surface structure of DiKoLAN AI was colored yellow, while the surface structure, i.e., the activities of the instructional action of both is the same. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

content knowledge (CK) by incorporating an additional dimension: sociocultural knowledge (SK). This expansion serves two primary purposes: first, to reflect the impact of sociocultural changes in the context of digitality (e.g., ethical considerations, changing communication and collaboration culture), and second, to emphasize the significance of sociocultural knowledge for STEM education and the development of effective teaching and learning processes. Sociocultural knowledge has been the focus of educational research for decades, particularly in the context of addressee orientation in teaching-learning processes. In light of the increasing heterogeneity in everyday teaching and changing social and cultural practices, this knowledge is becoming increasingly important in teacher training and professionalization.

The preceding sections have discussed the transformative impacts of AI on nearly all facets of society. This has led to the realization that the prevailing, predominantly technology-centric understanding of the TPACK model is no longer adequate to encompass all the relevant teaching dimensions of AI (cf. Mishra et al. [62]). For the DiKoLAN framework, this necessitates the development of a definition of the competencies of (prospective) teachers that is not exclusively based on technological knowledge. For instance, this cannot be solely achieved via the concepts of “Technological Knowledge (TK)”, “Technological Content Knowledge (TCK)”, or “Technological Pedagogical Knowledge (TPK)”. Instead, the interfaces to sociocultural knowledge (SK) must be considered to ensure the holistic didactic integration of digital technologies [66].

In the context of digitality, the culturally shaped human being becomes a central focus, while a purely technology-centered view, which dominated the early phases of school technology integration, is becoming increasingly less significant. For this reason, the DiKoLAN AI framework no longer relies exclusively on the TPACK approach. Instead, it has been modified to explicitly integrate digitality-related aspects in accordance with the DPACK model [66]. This integration embeds the knowledge essential for digitality – including digitality-related content knowledge, pedagogical professional knowledge, and subject-didactic professional knowledge – into teacher training.

This change in perspective is particularly relevant for the training of prospective teachers. It is important to acknowledge that the incorporation of sociocultural aspects cannot be solely assessed quantitatively, such as through an increase in explicitly articulated competence expectations. Rather, the sociocultural aspect is one of numerous factors that, when neglected, can have a substantial impact on the quality of teaching and learning processes—similar to insufficiently differentiated level-based planning.

The competence areas of DiKoLAN were previously addressed by the “Technical Core Competencies” and “Legal Framework”. However, in the emerging “age” of AI in education, these two areas are also affected by fundamental changes. In contrast to the situation with regard to the legal regulations concerning the previous integration of digital technology in the classroom, the legal framework and regulations for the use of AI are not yet advanced and differentiated. Consequently, educators must not only be aware of the potential of AI in and for the educational process, but also of potential dangers that may exist or arise when utilizing the technology properly, as these dangers may not yet be legally recognized. Given the absence of specific explicit legal regulations, such as the AI Act in the EU, it therefore seems appropriate that teachers should consciously carry out a risk assessment within the scope of their possibilities in the sense of a “risk assessment” already established in the natural sciences, particularly in instances where experience and guidelines are lacking. In order to adequately perform this task, as well as to leverage the technical capabilities of emerging technologies, it is critical that all teachers acquire basic computer science literacy in addition to purely technical skills (cf. Diethelm et al. [67]). This also applies to the subject-specific integration of basic IT skills [68–70].

Table 2
Genesis of DiKoLAN AI using selected examples of the competence area AI.DAP. Differences are highlighted in italics.

	DiKoLAN	DiKoLAN AI
Variant 1 Specification	DAP.T.D1 Describe didactic prerequisites of *digital* DAP for use in and effects on the respective teaching methods	AI.DAP.T.D1 Describe didactic requirements of *AI-based* DAP for use in and effects on the respective teaching methods
Variant 2 Augmentation	DAP.S.D2 Describe procedures (e.g., statistical programs, spreadsheets, databases) for • filtering, • calculations of new quantities, • preparation for visualization, • statistical analysis, • image, audio and video analysis, • linking of data, • automation in data processing.	AI.DAP.S.D1 Describe *AI-based* methods for • filtering, • calculations of new quantities, • preparation for visualization, • statistical analysis, • image, audio and video analysis, • linking of data, • automation in data processing, • *pattern recognition, and,* • *processing of large amounts of data.*
Variant 3 Redefinition	/	AI.DAP.S.N8 *Name the importance of the quality of the data and the resulting influence on the AI model.*
Variant 3 Deletion	DAP.S.D5 *Describe data structure of xml, csv files (also with semicolon separation).*	/

4.3. Development process of DiKoLAN AI and methods

A thorough examination of the eight individual competence areas of DiKoLAN, with a particular emphasis on AI, reveals three distinct approaches for determining and operationalizing competencies for DiKoLAN AI:

1. The first and simplest variant is the specification. This refers to general digital skills that can also be transferred to teaching with and about AI. This approach aligns with the methodology employed by Ning et al. [63] in the context of the AI-TPACK, wherein AI is regarded as a unifying specification of digital technologies in general, with a particular focus on the specific technology AI. In the area of competence formulation, this relationship becomes evident when the word “digital” is substituted with “AI-related” to describe the corresponding AI-related competence.
2. The second variant is augmentation. This is due, in part, to the extensive transformation processes occurring in society, science and education. In these contexts, the integration of AI technology necessitates a more profound examination. This entails, as articulated by Celik [65], the incorporation of ethical considerations and their impact on the educational process. The transformation of existing general digital competencies into specific AI-related competencies necessitates more than a mere exchange of words to describe suitable competence formulations, as new aspects also come into focus. The extended competencies were supplemented by expanded, precise formulations.
3. The third variant is characterized by redefinition or deletion. In the redefinition, new competencies that are not present in the general digital competencies of DiKoLAN were added. Examples of this are, in the sense of the DPACK model [66], skills in human-machine communication and collaboration, which historically hardly played a minor role in this form (i.e., without the use of AI). Conversely, general digital skills that have not been utilized in the context of AI-related skills have been eliminated (deletion). This deletion aligns with the approach of the AI-TPACK model by Ning et al. [63], which also stipulates the existence of general TC competencies other than AI-TC.

Table 2 shows the transformation of competencies using specific examples.

DiKoLAN AI was initially developed in three iterative cycles by the expert team of the Digital Core Competencies Working Group. Fig. 3 shows this iterative process.

4.3.1. First cycle (alpha-testing DiKoLAN AI)

The concept of adaptation was initiated by the nine members of the Digital Core Competencies Working Group from the DACH countries (Germany, Austria and Switzerland). These members come from the fields of Chemistry Education, Biology Education, Physics Education and Digital Education with a special focus on AI. The approach was subsequently presented to the scientific community from the DACH countries at the annual conference of the German Society of Chemistry and Physics Education [orig. Deutsche Gesellschaft für Didaktik der Chemie und Physik (GDCP)] in fall 2023, where it was discussed orally and feedback was obtained during a symposium about the procedure to generate DiKoLAN AI (see 4.3). This feedback contributed to the refinement of the approach. A fundamental starting point of DiKoLAN AI is the subject-specific design of the competencies. The formulation of the competencies was further refined by incorporating the findings from a parallel research paper by Berber et al. [17]. The initial version of the concrete formulation of the DiKoLAN AI competencies was then completed based on Berber et al. [17] and the feedback from the scientific community.

4.3.2. Second cycle (beta-testing DiKoLAN AI)

The resulting competency formulations, based on the approach previously described, were presented to a broad circle of experts from the field of AI in education from Germany, Austria, and Switzerland at a conference “AI in science education - perspectives, challenges & responsibility for research & teacher education” organized by the GDCP and the Joachim Herz Stiftung in February 2024. Subsequently, in a workshop, 24 experts from the fields of Chemistry Education, Biology Education, and Physics Education worked on the concrete formulations of the eight areas of competencies. The experts consisted of people from academia, such as university professors, postdocs, doctoral students and people from educational practice, such as active teachers with teaching experience. The formulation process involved the use of large posters, with each group consisting of three experts. The posters contained all the formulated DiKoLAN AI competencies in the Track Changes mode, so it was clear which changes were made based on the procedure (see 4.3.1) for formulating the AI competences. Subsequently, the experts contributed comments, suggestions for improvement, and feedback, which were then discussed by the group in the next step. The expert groups were explicitly instructed to examine the specific formulations critically and to mark formulations that were agreed with a symbol for agreement. In instances were a formulation was not approved or was viewed critically, the expert teams were instructed to either delete words or modify formulations, contribute their own formulations, or

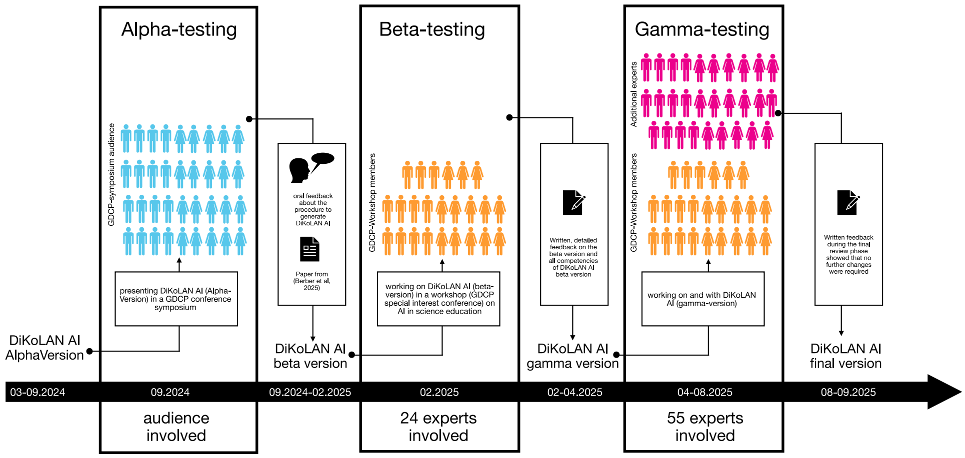

Fig. 3. Iterative process of creating DiKoLAN AI and involving experts in the development process.

introduce new or additional competence expectations. Additionally, the experts were also asked whether they considered the DiKoLAN framework to be suitable for specifying AI-related competencies. The overall feedback resulted in minor suggestions for modifications to the specific formulation of the individual competencies of the eight competence areas. These were distributed roughly equally across the eight competence areas. The experts also evaluated the procedure for defining AI-specific competencies based on DiKoLAN with a questionnaire, rating it as very suitable.

4.3.3. Third cycle (gamma-testing DiKoLAN AI)

The results of the third phase were subsequently addressed and incorporated by the Digital Core Competencies Working Group. The revised framework was then made available for review in April 2024 to a total of 55 experts (additional to the nine experts from the *Digital Core Competencies Working Group* from Germany, Austria, and Switzerland in the fields of biology education, chemistry education, physics education, computer Science education, digital educational science, computational linguistics, and media technology. Participants were tasked with the provision of exemplars of teaching with and about AI, with the DiKoLAN AI framework serving as a reference point. This step also aimed to evaluate the practical applicability of the DiKoLAN AI framework. As no further suggestions for changes were received during testing and working with the gamma version, the gamma version was adopted as the finale version, along with minor spelling corrections. The result of this process was the creation of a book intended for German-speaking audiences. This book includes the DiKoLAN AI framework, as well as 20 specific implementation examples [71]. In this publication, we outline the framework in a manner analogous to the original DiKoLAN framework [12] and discuss its efficacy in practice. In order to get an overview of the efficacy in practice, we have also prepared a heat map for the 20 case studies.

5. The DiKoLAN AI framework

5.1. Assessment, feedback and adaptivity (AI.AFA)

With the growing demand for adaptive teaching and the associated formative assessment for the individual promotion of teaching and learning processes, activities and skills in this area are becoming more and more important. Teachers are increasingly challenged by the associated tasks, also in the context of feedback. The provision of feedback necessitates the implementation of learning status diagnoses, subsequent analyses and the development of individual support measures and learning concepts. Technologies that alleviate the workload of teachers by intelligently and adaptively providing diagnostic tasks, evaluating them, and selecting appropriate learning and practice tasks [72], and the competent use of such systems will be essential for successful adaptive teaching in the future. Consequently, it is necessary that teachers possess a comprehensive understanding of existing, suitable systems, their functionality, as well as the skills to operate and integrate them into the classroom.

It is also key to understand AI-based technologies and their context in which they are used for assessment, such as the diagnosis of student perceptions, the (partially) automated provision of feedback in appropriate formats including adaptive suggestions for further learning design, and micro-level support. Peer feedback and self-comparison with other learners using AI-based technologies can enhance learning processes while complying with data protection aspects [73]. It is essential that (prospective) teachers are aware of the structure and format of the data collected and stored by AI. On the one hand, this awareness enables the assessment and estimation of the necessary security requirements. On the other hand, it is essential to recognize that AI may not be accepted by learners for socio-cultural reasons or may even act in a discriminatory manner [74].

In addition, it will be useful for teachers to be able to address how AI is used in the subject areas of assessment, feedback and adaptivity. To illustrate, a large language model can be utilized to assist researchers in their role of group supervisors or in the process of experimental design. This technology can function not only as a feedback mechanism for research steps or implementation plans, but also to structure an entire laboratory. With the AI system Coscientist [23], an autonomous laboratory is created that can plan and carry out series of experiments based on text input from researchers. This process encompasses the planning and execution of experiments across multiple stations, with real-time optimization through feedback loops based on the outcomes of these experiments.

Similarly, if teachers are aware of the potential and possibilities of regular formative assessment, they can use AI systems to continuously monitor and track the learning status of learners. This enables individual learning paths and support, which often cannot be achieved with traditional methods, even with intensive use of resources [72]. However, it is critical to acknowledge the implications of sociocultural factors, particularly the acceptance of these systems by learners and the potential for cultural bias inherent in AI algorithms [74].

When used competently, AI allows materials to be differentiated or individualized in a time-saving manner at the macro level of adaptation [75]. At the micro level, AI enables timely, accurate and individualized feedback, for example through the use of TutorBots that provide specific feedback to learners [8,76]. Teachers should also be informed about the potential impacts and risks of using AI within educational settings [56]. Data protection and the potential perpetuation of social

Table 3
Competencies in the area of *Assessment, Feedback and Adaptivity* (AI.AFA).

Competence level	Teaching (TPACK)	Methods, Digitality (TPK)	Content-specific context (TCK)	Special Tools (TK)
Name	AI.AFA.T.N1: Name conditions and scenarios for the beneficial use of AI-based feedback, tutoring and assessment systems in teaching-learning situations. AI.AFA.T.N2: Name context-specific possibilities for AI-based methods in the planning and implementation of differentiated to adaptive teaching.	AI.AFA.M.N1: Name potentials of AI-based forms of learning assistance (synchronous or asynchronous), e.g., by means of • digital feedback via comments etc., • digital glossaries and FAQ-like support. AI.AFA.M.N2: Name advantages and disadvantages of AI-based digital implementations in dealing with learning products for digital learning support, e.g., in various modes of (social) interaction. AI.AFA.M.N3: Name potentials of diagnoses and summative and formative feedback made possible by AI-based tools. AI.AFA.M.N4: Name potentials for self-directed learning with AI-based tutorial systems incl. assessment systems. AI.AFA.M.N5: Name data, parameters and indicators that can be collected through AI-based monitoring and used and communicated for the discussion of learning activities/processes. AI.AFA.M.N6: Name differences in the design of AI-generated and technology-supported feedback by the teacher.	AI.AFA.C.N1: Name AI-based technologies for controlling scientific research documentation as an example of support systems with feedback (e.g., AI-based research data management systems and Large Language Models (LLMs)). AI.AFA.C.N2: Name AI-based assessments to control workflows and processes (e.g., real-time process optimization).	AI.AFA.S.N1: Name AI-based technologies that offer functionalities in the area of, e.g.: • structuring study schedules in terms of content and time to suit learning groups, • giving simple (i.e., "right-or-wrong") feedback, • giving more differentiated (i.e., elaborated) feedback, • creating and integrating summative assessments, • recording and documenting learning progress (formative assessment), • audience response systems, such as survey systems, mind or concept mapping tools (automated live evaluation). AI.AFA.S.N2: Name AI-based technologies for assessment, feedback, diagnosis and their adaptive design that can (partially) automatically support learning support with the above-mentioned functions. AI.AFA.S.N3: Name AI-based technologies that enable anonymous, individual comparison with and/or in the learning groups. AI.AFA.S.N4: Name technical possibilities for implementation of AI-based audio and video feedback.
Describe (including necessary procedures)	AI.AFA.T.D1: Describe didactically justified procedures for the appropriate use of AI-based assessment and tutoring technologies in specialist contexts.	AI.AFA.M.D1: Describe the advantages, disadvantages and limitations of AI-based feedback technologies, tutoring technologies, and assessment/diagnosis technologies. AI.AFA.M.D2: Describe pedagogical requirements for the methodical use of AI-based feedback, tutoring and assessment/diagnosis technologies, e.g., with regard to: • expenditure of time, • organizational forms and supervision, • self-directed, individual learning, • interest, motivation, and cultural contexts. AI.AFA.M.D3: Describe strategies for using AI-based feedback, tutoring and assessment/diagnosis technologies in teaching-learning processes.		AI.AFA.S.D1: Describe data, e.g., for registration and/or use, which the above-mentioned AI-based technologies collect and store in order to be able to act securely with regard to the General Data Protection Regulation (GDPR). AI.AFA.S.D2: Describe technical requirements that are necessary for specific AI-based technologies for use in assessment, feedback, and diagnosis. AI.AFA.S.D3: Describe, from a technical perspective, the implementation and associated workflows and processes for selecting and utilizing the AI-based technologies mentioned in AI.AFA.S.N1 in accordance with the given technical and cultural contexts and conditions of the digital infrastructure.
Use/ Apply (practical and functional realization)	AI.AFA.T.A1: Planning and implementation of teaching-learning processes (i.e., in teaching scenarios and/or beyond) involving AI-based technologies for assessment, feedback, diagnosis and their adaptive use.			

discrimination are among other important critical issues that must be addressed [77].

Formative assessment must to address didactically relevant aspects (e.g., necessary basics, intended learning objectives) with precision and through appropriate integration in the classroom. Depending on the assessment's purpose, appropriate systems must be selected. The selection of systems should be based on the results of adaptive and differentiating approaches. It should also be possible to use supporting technologies (e.g., tutoring and practice systems) for further learning processes. Suitable technological assistance for digitally implemented supporting concepts that are coordinated in terms of subject-specific didactics must be selected in accordance with the teaching context and the objective. AI-based systems possess distinct capabilities and functionalities in this regard when compared to non-AI-based systems.

> **The competence area Assessment, Feedback and Adaptivity (AI.AFA)** comprises the individual ability to use AI-based tools systematically for feedback, tutorial support, and assessment. This also includes the selection of AI-based systems to support and evaluate learners, including the provision of system-guided and/or technology-enhanced feedback, individual support and differentiation as well as performance assessment, as a central facet of learning support. In all cases, digitality-related aspects must also be given greater consideration.

Table 3 summarizes practice-oriented suggestions for AI-related competencies in the area of *Assessment, Feedback and Adaptivity*.

5.2. Documentation (AI.DOC)

Learning and development processes are inconceivable without documentation. Whether it is the necessary data to derive a fact or the documentation of a process, without complete and comprehensible documentation, no processes can be traced or checked. Documentation is therefore of crucial importance in all teaching activities, but also in participation processes, e.g. when implementing digital technologies and methods in teaching and learning processes. The data required for this can consist of various types of data and can therefore be produced using different AI-based technologies. Competent teachers are aware, for example, of the problem that when creating images and videos, AI-supported algorithms change the color values due to the identification of objects in such a way that they no longer correspond to the actually observed (real) ones. This, in turn, makes learners aware of the possible dangers in the subsequent analysis of this data, for example, for colorimetry.

For example, AI-supported dictation and transcription services can be used for documentation. Whether recorded with a mobile device or dictated directly into the web browser, speech recognition is used to automatically convert spoken input into written text. It is important to be aware that the processed data can also be saved for further training of AI models. Teachers must comply with the applicable laws on the handling and processing of personal data. Backups can be created and monitored automatically using a combination of generative AI and conventional ML models. The AI-powered technology detects backup failures, schedules and performs new backups, and generates comprehensive reports outlining the necessary next steps. Since teachers manage sensitive personal data, it is necessary to make them aware of the benefits and potential risks, particularly in terms of data security and with AI-generated backups and the use of public chatbot services. Backups can be created automatically using ML models while avoiding the potentially insecure transfer of data to third-party providers.

In addition to the analysis and processing of data, the storage of data plays an important role in the specialist sciences. The utilization of AI in this domain is not merely a potential solution but a definitive prospect, capable of either supplementing or entirely assuming this function. One such approach in catalysis chemistry involves the use of ML (specifically, a global neural network, GNN) to automatically generate a reaction database. A secondary AI model can then extract all information required to predict the reactions possible in a heterogeneous catalysis system from this database [41]. AI is also able to take over previously analog or manual work steps in the area of data storage; thanks to the continuous improvement of natural language processing (NLP), virtual, AI-based laboratory language assistants can understand and implement work orders – such as the documentation of measured values and observations – with high accuracy [e.g., 42,43].

Before using AI-based documentation in teaching and learning scenarios, it is essential that teachers check whether the software used is accessible to all participants without restriction and whether it is available centrally or on the devices used. In BYOD scenarios, the social and personal ramifications of the latter, in particular, must be given full consideration, as only a select group of people have access to the devices that meet the very high hardware requirements.

The use in teaching-learning scenarios should be designed to convey both a basic understanding of the functionalities as well as the possibilities and limitations of AI-based documentation and AI-supported backups. This approach entails the incorporation of appropriate and professional learning experiences with, through, and about AI-based technology in science lessons. To address the social and personal consequences of use, the development of suitable organizational and social forms are required. These frameworks should facilitate the acquisition of essential skills for responsible AI utilization while concurrently mitigating exclusion due to access barriers, restrictions, or inadequate hardware. Speech recognition is a process that converts spoken language into written text, thereby facilitating data processing via AI-based documentation. Consequently, AI-based documentation is intricately linked to AI-based data processing, encompassing the competencies that must be considered for ensuring compliance with the General Data Protection Regulation (GDPR). Given the increase in the use of digital technologies in the development process, learners need to be made aware of the need to properly create and check data backups. With the integration of backups based on ML models, learners experience the advantages and risks of these technologies, enabling them to select a suitable technology or method in comparable situations.

> **The competence area Documentation (AI.DOC)** describes the individual ability to use AI-based tools for the systematic filing and permanent storage of data and information in order to use them appropriately. This also includes taking, editing and integrating photos, images and videos using AI-based functions, combining and storing different media, saving and archiving information in a structured manner and presenting processes and contexts. This also includes the ability to reflect on the use of AI in documentation as well as in-depth media criticism of documented data. In each case, digitality-related aspects must also be given greater consideration.

The practical competency expectations for AI-based *Documentation* listed in Table 4 clearly show close links to more subject-specific competency areas, such as data acquisition, and data processing.

5.3. Presentation (AI.PRE)

It is difficult to imagine learning and teaching that does not incorporate visual elements in the dissemination of knowledge. Digitalization has profoundly transformed and expanded the scope of visualization in the teaching-learning process. A comparable expansion is anticipated with the introduction of AI in the field of presentation and visualization. AI-supported tools have the potential to not only facilitate the visualization of voluminous and intricate data sets but also to transform the landscape of education and research. Moreover, they can also be used to predict and present 3D-structures that were previously unattainable [24,78]. AI-powered tools can also help users to visualize ideas in a graphically appealing way, thereby enhancing creativity and reducing the time required for this process. At the same time, however, there is also a risk of misuse of this new technology, as AI-supported tools can be used to create or modify existing and new visualizations. It is necessary for the future of teaching and learning processes to develop a comprehensive understanding of the potential of AI-supported tools to expand individual technical capabilities and their application in the domain of presentation media. Given the transformative nature of these technologies, the extent of this change must be carefully considered from both technical and didactic perspectives.

AI-supported visualizations have also been incorporated into scientific research, including the creation of overviews using generative AI and also the automated graphical processing of data as diagrams or 3D-models. For instance, AI has the capacity to predict the folding of any amino acid sequence [24,78] by calculating the most likely 3D-structure and displaying it as a three-dimensional molecule. The use of color in these visualizations serves to denote the reliability of the prediction in the different regions. In a similar manner, models (e.g., of laboratory equipment) can be created using AI and subsequently printed using a 3D-printer [32].

When using and creating AI-based presentation media in teaching and learning scenarios, the laws, rules, and guidelines to be followed should be made transparent. The impact on teaching and learning as well as media-critical aspects must also be considered. The integration of AI-supported presentation tools within science education can support appropriate and professional learning with, through, and about AI-based technology. It is particularly important that teachers develop learners' competencies in the application options when presenting science content.

> **The competence area Presentation (AI.PRE)** describes the individual ability to use digital media for the cognitive and communication process in a target- and addressee-oriented manner, as well as knowledge about the limits and potential of different digital presentation media. This also includes the ability to reflect on the use of AI in the creation of presentation media as well as in-depth media criticism. In each case, digitality-related aspects must also be given greater consideration.

Table 5 summarizes the expectations in the competency area *Presentation* for science teachers.

5.4. Communication/collaboration (AI.COM)

The act of teaching depends on communication and the exchange and management of information. In accordance with social constructivist learning theories, communication and information promote participation in competence-oriented, technology-supported learning [79]. Therefore, the integration of suitable communication techniques and phases as well as collaborative elements for participation in the learning process, are key elements of teaching. The integration of AI-based technology in the classroom and beyond has given rise to a variety of new forms of communication and collaboration. What all forms have in common is that, in addition to supporting human–human

Table 4
Competencies in the area of *Documentation* (AI.DOC).

Competence level	Teaching (TPACK)	Methods, Digitality (TPK)	Content-specific context (TCK)	Special Tools (TK)
Name	AI.DOC.T.N1: Name conditions and scenarios for the appropriate use of AI-based documentation in teaching-learning scenarios.	AI.DOC.M.N1: Name methodological and socio-cultural aspects that can be relevant when using digital documentation with AI software in the classroom, e.g., • access to the AI software, • expenditure of time, • hardware requirements, • access restrictions, • socio-cultural and personal impact of the documentation of the person, • implications of the (training) data for the AI software.	AI.DOC.C.N1: Name possibilities of appropriate AI-based documentation/versioning and data archiving (e.g., creation of a database on chemical reactions/genes using a "Global Neural Network"), taking into account the citation rules. AI.DOC.C.N2: Name methods of digital data documentation with AI software in research scenarios (e.g., AI-based image documentation, storage of experimental data and protocols with a "laboratory language assistant").	AI.DOC.S.N1: Name technical approaches, such as: • possibilities for AI-based digital documentation of, e.g., protocols, experiments, data, analysis processes, digital herbaria, • options for version management and file archiving with AI-based technology.
Describe (including necessary procedures)	AI.DOC.T.D1: Describe didactically justified scenarios for the appropriate use of AI-based digital documentation/versioning or data archiving/backup creation for specific teaching/learning scenarios.	AI.DOC.M.D1: Describe methodological advantages and disadvantages as well as limitations of AI-based documentation in relation to teaching-learning scenarios.	AI.DOC.C.D1: Describe methods of digital data documentation with AI software in research scenarios (e.g., AI-based image documentation, storage of experimental data and protocols with a "laboratory language assistant").	AI.DOC.S.D1: Describe the documentation options listed under AI.DOC.S.N1 with regard to: • existing functions, • technical conditions, • technical requirements, • technical advantages and disadvantages (e.g., automated backups with AI software).
Use/Apply (practical and functional realization)	AI.DOC.T.A1: Planning of complete teaching scenarios with professional application of AI-based documentation/versioning or data archiving/backup creation, taking into account suitable social organization of the classroom and modes of (social) interaction (e.g., long-term documentation over the course of upper secondary education [ISCED 3]). AI.DOC.T.A2: Implementation of complete teaching scenarios with professional application of AI-based documentation/versioning or data archiving/backup creation, taking into account suitable social organization of the classroom and modes of (social) interaction (e.g., long-term documentation over the course of upper secondary education [ISCED 3]).			

communication and collaboration, a new form of communication and collaboration is emerging: that of a human to a machine, without the need for a second person.

The enhancement or support of collaborative processes, such as text and data processing, is a potential benefit of AI-based technology. For example, AI chatbots can function as collaboration partners by verifying formulas during data processing or providing assistance during evaluation. The transmission of information, in the sense of communication, can take place via voice, text or images, among other things. In all these forms, AI-based technology can support communication between people through simultaneous translation. Similarly, almost any conversation with machines, such as AI chatbots, can be initiated without the prerequisite of a pre-programmed dialogue. In such interactions, specific linguistic instructions (prompts) become important in order to receive the desired response.

AI-supported simultaneous translation offers an inclusive means of enhancing global accessibility to scientific research by removing language barriers and facilitating international collaboration. However, effective use of such tools – as well as other generative AI systems – requires specific skills in prompt engineering [80]. Moreover, the accuracy of generated content, including technical details and data-based information, must be critically assessed to prevent misunderstandings that could compromise research quality. Interactions with AI systems often involve the storage and processing of user input for training purposes, raising concerns about data privacy. Additionally, AI systems may produce so-called *hallucinations* – fabricated or inaccurate outputs – which can disrupt communication and collaboration. Nevertheless, AI chatbots can also serve as tools for perspective-taking or as conversational partners, such as in foreign language learning. As interactions with machines increasingly replace human-to-human communication, there may be broader implications for interpersonal communication and socio-cultural behavior. These potentials – as well as the limitations and ethical considerations – should be addressed explicitly in science education. By critically engaging with AI technologies, students can develop meaningful competencies in subject-specific human-machine communication. Teachers play a crucial role in fostering these competencies to support professional and reflective learning with, through, and about AI.

> **The competence area Communication and Collaboration (AI.COM)** comprises the individual ability to plan synchronous or asynchronous work of individuals or groups towards a common objective using digital AI-based tools and to carry it out with learners. To this end, joint files or products are created, edited and communicated using AI-based tools. As an extension to the previous understanding of communication and collaboration (person-to-person), the facet of person-to-machine communication and collaboration is added. Digitality-related aspects must also be increasingly taken into account.

Table 6 summarizes practical suggestions for AI-based competencies in the area of *Communication and Collaboration*.

5.5. Information search and evaluation (AI.ISE)

As AI undergoes progressive development, its influence on research services is also increasing considerably. In order to address these emerging requirements, the competence area of information search and evaluation has been extended to include AI-related competencies. This

Table 5
Competencies in the area of *Presentation* (AI.PRE).

Competence level	Teaching (TPACK)	Methods, Digitality (TPK)	Content-specific context (TCK)	Special Tools (TK)
Name	AI.PRE.T.N1: Name suitable AI-based alternatives to (scientific) presentation media for use in schools. AI.PRE.T.N2: Name different scenarios for the appropriate use of AI-based presentation media for specific teaching/learning settings/contexts (appropriate to the addressee, subject and target).	AI.PRE.M.N1: Name principles/criteria for designing digital presentation media appropriate for the target audience with AI-based technology. AI.PRE.M.N2: Name possible aspects that can be affected by the use of AI-based digital presentation media in learning and teaching, e.g., with regard to • expenditure of time, • forms of organization, • form of presentation, • methods, • media knowledge/familiarization, • interest and motivation, • personal and social consequences.	AI.PRE.C.N1: Name several scientific scenarios and, if applicable, contexts for AI-based • forms of presentation (e.g., summaries of research, literature), • presentation of process (e.g., visualization of data), • presentation software that meets current scientific requirements.	AI.PRE.S.N1: Name several technical options for the AI-based creation of presentation media, e.g., of • photorealistic or illustrative images, • time-based visual media, • auditory media, • presentation slides, • websites.
Describe (including necessary procedures)	AI.PRE.T.D1: Describe didactic prerequisites for the use of AI-based presentation media in the classroom, their impact on the respective teaching methods, and the access to basic competencies (especially the competency area of communication) enabled by these systems, especially in the context of inclusive teaching and learning.		AI.PRE.C.D1: Describe selected scientific AI-based scenarios, e.g., • forms of presentation, • presentation of processes.	AI.PRE.S.D1: Describe at least one way of technical implementation for each type of presentation, including the necessary procedure with reference to current AI-based technologies. AI.PRE.S.D2: Describe the features/functionalities, technical requirements and any limitations of the respective AI software.
Use/Apply (practical and functional realization)	AI.PRE.T.A1: Planning of complete teaching scenarios with the integration of AI-based presentation media and forms and the consideration of suitable social organization of the classroom and modes of (social) interaction. AI.PRE.T.A2: Implementation of complete teaching scenarios with the integration of AI-based presentation media and forms and the consideration of suitable social organization of the classroom and modes of (social) interaction.		AI.PRE.C.A1: Creation and demonstration of presentations in a scientific context using AI-based presentation media, e.g., • creation of models for the production of laboratory equipment using 3D printers.	AI.PRE.S.A1: Performing commissioning, calibration and usage for at least one example of each type of AI-based presentation capability listed above.

Table 6
Competencies in the area of *Communication and Collaboration* (AI.COM).

Competence level	Teaching (TPACK)	Methods, Digitality (TPK)	Content-specific context (TCK)	Special Tools (TK)
Name	AI.COM.T.N1: Name technologies that are suitable for appropriate AI-based communication and collaboration in a specific teaching-learning scenario.	AI.COM.M.N1: Name possible limits and impacts/aspects of the respective AI-based software use in the classroom with regard to • communication beyond the classroom with foreign language speakers, • communication with AI-based systems (assistants), • technical problems and preparation time, • self-organization and self-control, • data security (use of input as training data).	AI.COM.C.N1: Name ways of communicating with international colleagues using suitable AI-based systems (e.g., offering simultaneous translation). AI.COM.C.N2: Name subject-specific options for communicating with AI-based systems, e.g., prompts for subject-specific language models for simultaneous translations.	AI.COM.S.N1: Name AI-based software for collaborative text and data processing. AI.COM.S.N2: Name possibilities for communication with AI-based software, e.g., via • language, • text, • image.
Describe (including necessary procedures)	AI.COM.T.D1: Describe application scenarios of AI-based technology and suitable strategies for using it in the communication process. AI.COM.T.D2: Describe collaboration scenarios for the use of AI-based technology. AI.COM.T.D3: Describe didactic prerequisites for the use of AI-based technology in the classroom, the effects of this on the respective teaching methods and the possible access to basic skills (especially the competence area of communication) and in inclusive teaching and learning.	AI.COM.M.D1: Describe the advantages and disadvantages of using AI-based software in the classroom with regard to the aspects mentioned above.	AI.COM.C.D1: Describe advantages of AI-based communication with international colleagues for research and individual projects. AI.COM.C.D2: Describe advantages and limitations of communication/collaboration with AI-based systems for research and individual projects.	AI.COM.S.D1: Describe the AI-based software mentioned under AI.COM.S.N1 with regard to its application. AI.COM.S.D2: Describe the AI-based software mentioned under AI.COM.S.N2 with regard to its application.
Use/Apply (practical and functional realization)	AI.COM.T.A1: Planning of complete teaching scenarios with appropriate use of AI-based technology, taking into account suitable social organization of the classroom and modes of (social) interaction. AI.COM.T.A2: Implementation of complete teaching scenarios with appropriate use of AI-based technology, taking into account suitable social organization of the classroom and modes of (social) interaction. AI.COM.T.A3: Introduction of learners to AI-based communication and collaboration technologies. AI.COM.T.A4: Instruction of learners in subject-specific communication with AI-based technologies (human-machine communication).			AI.COM.S.A1: Use AI-based collaborative software for text and data processing. AI.COM.S.A2: Use AI-based systems for collaborative data management. AI.COM.S.A3: Communicate with AI-based systems using the options mentioned in AI.COM.S.N2.

adaptation is characterized, among other things, by AI-supported recommendation services that use algorithms to decide what information to display to users [81]. These entities employ training data, textual content, metadata, and click data to filter results, thereby creating a self-reinforcing cycle that leads to an increase in bias and the formation of filter bubbles. This phenomenon is not exclusive to specialized search assistants but also to content filtering by corporations and the entertainment industry. It is critical for teachers to take this into account when researching and evaluating online sources, and to develop a technical understanding of AI search engines and recommendation systems. In the classroom, the necessity of AI skills is evident as teachers pass on their understanding to students who are expected to become informed and critical users within a society characterized by disinformation and distorted facts [82].

AI-based research utilizes algorithms to extract relevant information from voluminous data sets, with the precision of the results highly dependent on the quality of the training data, the specificity of the prompts, and the accessible data sources. The effectiveness and limitations of these technologies are influenced by factors such as data bias, access to comprehensive databases and the ability of AI to detect misinformation. The quality of AI research results is assessed by criteria such as timeliness, scientific validity and the quality of the references used. The underlying training data often determines both the strengths and weaknesses of the results.

The automation of extensive substance and protein databases, as well as scientific literature searches, can be achieved through the usage of various AI models. These systems have been shown to enhance efficiency when compared to non-AI-driven methods [e.g., 22,31]. For example, a targeted AI-based literature search can be used to extract synthesis conditions [32] or to determine and optimize experimental procedures [23]. Furthermore, these systems possess the capability to evaluate the retrieved information (e.g., the binding capacity of a substance to an enzyme; [e.g., 25]). This capacity to assess the reliability of the retrieved information is a significant contribution of these systems. These processes have the potential to facilitate the application of AI-supported literature analysis in predicting future research priorities [22]. As in everyday use, when using generative AI systems for scientific purposes, it is important to ensure that the information retrieved is technically correct and up to date. The effective and appropriate use of these research functions by means of corresponding prompts is also necessary [80].

AI-supported research tools and digital databases can significantly facilitate information retrieval in teaching and learning scenarios by providing quick access to a variety of resources. However, it is important to critically evaluate the quality and reliability of the information to avoid over-reliance on technology and loss of critical thinking skills. When planning and implementing teaching units that address the use of AI-supported research tools, particular attention should be paid to promoting research and evaluation competencies and reflecting on ethical aspects. These approaches contribute to a deeper understanding of the potential and limitations of AI in educational contexts. AI-based research in subject lessons requires careful planning and implementation in order to meet (subject) didactic and ethical standards. Teachers should tailor the use of AI in teaching-learning scenarios precisely to promote critical thinking and subject-specific didactic objectives. The evaluation of research results should be based on criteria such as accuracy and objectivity. Students should be guided to critically question the credibility of (subject-specific) information. It is crucial that the integration of AI technology into subject lessons be transparent, thereby fostering an environment conducive to critical reflection.

> **The competence area Information Search and Evaluation (AI.ISE)** includes technical and individual competencies to use digital tools for AI-based research, to adjust them appropriately and to evaluate their results qualitatively. Digitality-related aspects must also need to be taken into account to a greater extent.

Table 7 summarizes the expectations of teachers in the competency area *Information Search and Evaluation*.

5.6. *Data acquisition (AI.DAQ)*

Due to the increasing availability of powerful AI algorithms, an increasing number of data acquisition systems are incorporating AI-supported options for the pre-processing and management of measurement data. Although the application of AI algorithms is primarily concerns data processing, their integration also influences teaching activities in the field of measurement and data acquisition, thereby necessitating specific competencies in this domain. With the adoption of AI technologies, the distinction between data acquisition and data processing shifts from purely technical differentiation to a human-centered perspective. Ultimately, it is not the technical procedure themselves that are decisive, but rather the implications and benefits of their application for individuals.

Accordingly, it is also essential to consider the broader aspects of digitality associated with AI-supported data acquisition. AI can streamline the recording of measured values and make it possible to record and manage large volumes of data. Moreover, AI is now opening up entirely new fields of data acquisition, beyond what is currently emphasized in frameworks such as DiKoLAN, which already places greater focus on digital data acquisition rather than mere digital data collection. While DiKoLAN accounts for data extraction from images and audio material, AI significantly expands these capabilities and introduces new requirements for teacher competencies. As the complexity and extend of data processing increases (especially in the pre-processing of (measurement) data), certain elements of data quality may become difficult to verify. As a result, both learners and teachers must be made aware to the uncertainties that can arise when AI is involved. A prominent example of “measurement results” falsified by AI is that Samsung smartphones were able to take detailed photos of the moon, even when a similar object, not the moon, was photographed [83]. Competent teachers must not only recognize such risks but also guide students in critically evaluating the implications of unreflective use of AI-based measurement and data acquisition technologies.

The use of AI-based measurement and data acquisition requires a solid understanding of the technologies used. This includes knowledge of AI-based sensor technologies, intelligent IoT (Internet of Things) devices and the functioning of AI algorithms for pre-processing collected data. Competencies in this area include the ability to select and integrate suitable hardware and software for data acquisition and to implement AI algorithms for initial automated data analysis. In biology lessons, for instance, the use of AI-based apps to identify animals and plants from recorded images is already widespread. Similarly, bird calls can be identified from audio recordings using AI. The number of certain objects in images can also be counted or relationships between objects recognized. Such AI-based recognition not only supports but also simplify non-digital processes, such as species identification and the estimation of population density. However, the intelligent pre-processing of data is also brings increased technical demands on measurement and data acquisition systems. As a result, careful consideration of the technical requirements becomes even more important due to the requirements of intelligent pre-processing.

AI-based measurement and data acquisition also plays an significant role across the scientific disciplines. In biological and biomedical research, for example, the AI-supported extraction of data from image material has become an essential method. In particular, machine learning and deep learning (DL) approaches have proven highly effective in the classification and evaluation of microscopic or X-ray images, as they are capable of recognizing and categorizing complex patterns and fine-grained details [18]. For example, DL-based systems are widely employed in the analysis and interpretation of microscopic images of histological sections of various tissues [19,20]. Among other things, this enables the automatic and highly accurate detection and diagnosis of lung cancer [19]. In the field of microbiology, AI is used, for example, to interpret bacterial culture plates [21]. Using Convolutional Neural Network (CNNs), bacterial colonies of culture plates are automatically

Table 7
Competencies in the area of *Information Search and Evaluation* (AI.ISE).

Competence level	Teaching (TPACK)	Methods, Digitality (TPK)	Content-specific context (TCK)	Special Tools (TK)
Name	AI.ISE.T.N1: Name conditions and scenarios for the appropriate use of AI-based research in teaching-learning scenarios. AAI.ISE.T.N2: Name criteria for evaluating the results of AI-based research in the classroom.	AI.ISE.M.N1: Name the advantages, disadvantages and limitations of AI-based research for use in teaching and learning scenarios.	AI.ISE.C.N1: Name AI-based technologies for searching in science-specific databases, e.g., • Large Language Models, • Tensor Networks, • Graph-convolution Neural Networks. AI.ISE.C.N2: Name subject-specific contexts for AI research or in the field of data research: • presentation of current research priorities, • summary of information from research literature, • identification of substances with specific properties from databases. AI.ISE.C.N3: Name criteria for evaluating AI-based research in a scientific context, e.g., • timeliness of the information, • technical correctness, • scope/style/design, • data basis, • recognizability of author and references. AI.ISE.C.N4: Name subject-specific contexts in which AI-based assessments are carried out, e.g., • binding capacity of ligands to target structures.	AI.ISE.S.N1: Name possibilities of AI-based research. AI.ISE.S.N2: Name the prompts required to carry out AI-based searches. AI.ISE.S.N3: Name factors that influence the limitations of AI in the evaluation of information (understanding, bias, probability of incorrect information), such as data basis, access to databases, training data. AI.ISE.S.N4: Name limitations of the results resulting from the training or reference data sets. AI.ISE.S.N5: Name factors influencing the search result when using AI systems: Basic functioning of the algorithm used, e.g., access options/depth of processing of the AI system to/from data.
Describe (including necessary procedures)	AI.ISE.T.D1: Describe appropriate application scenarios for AI-based research and how to evaluate the results based on criteria.	AI.ISE.M.D1: Describe the advantages, disadvantages and limitations of digital databases and search engines for AI-based research when used in teaching and learning scenarios.	AI.ISE.C.D1: Describe prompting strategies for an AI-based scientific search query. AI.ISE.C.D2: Describe the evaluation of search results found with AI-based technology using an example. AI.ISE.C.D3: Describe the factors influencing search results when using AI-based technology using an example. AI.ISE.C.D4: Describe at least two of the criteria listed under AI.ISE.C.N3.	AI.ISE.S.D1: Describe important settings and instructions for processing AI-based research tasks using an example. AI.ISE.S.D2: Describe criteria for evaluating the quality of AI-based search results, e.g., • timeliness, • scientificity, • references. AI.ISE.S.D3: Describe the influence of limitations resulting from training or reference data sets on the results.
Use/Apply (practical and functional realization)	AI.ISE.T.A1: Planning of complete teaching scenarios with the integration of AI-based research as well as the evaluation of the results based on quality criteria and the consideration of suitable social organization of the classroom and modes of (social) interaction. AI.ISE.T.A2: Implementation of complete teaching-learning scenarios with the integration of AI-based research as well as the evaluation of the results based on quality criteria and the consideration of suitable social organization of the classroom and modes of (social) interaction.	AI.ISE.M.A1: Planning and implementation of teaching scenarios in which the (subject-independent) use of AI-based research software is addressed.	AI.ISE.C.A1: Carry out subject-specific research using AI-based technology and evaluate the results.	AI.ISE.S.A1: Carry out a search with an AI-based system and evaluate the results.

counted and the respective species identified in order to be able to draw conclusions about diseases or contamination of the smear site. The automated recording of measured values through specialized AI systems within autonomous laboratories represents another scientifically relevant application of AI in research [23]. These autonomous laboratories mark an innovative development in chemistry by combining AI with automated experimental techniques, enabling the rapid and efficient resolution of complex chemical problems. Large language models (LLMs) also play a central role in these setups, supporting experimental planning, data interpretation, and the optimization of laboratory workflows [23]. Examples such as AlphaFlow and CoScientist illustrate how AI-based functionalities can augment human research capabilities by delegating specific laboratory tasks to specialized AI-powered assistants [23,44]. AI can also be used to collect secondary data from raw data; Xing et al. [84] show how AI-supported electrochemical measurements can be used to determine the concentrations of individual substances in complex mixtures. In physics, AI is employed, for example, to analyze heavy-ion collisions by extracting information about the number and type of particles involved directly from the detector data [85].

However, the application of AI-supported measurement and data acquisition also raises important ethical, legal and social requirements and implications. This includes issues of privacy policies, data security, fairness and transparency in data collection. Competencies in this area therefore encompass not only technical proficiency but also the ability to critically reflect on the implications of AI for data protection, particularly with regards to the safeguarding of student data, and to foster responsible and informed use of the collected data. The implementation of AI-supported measurement and data collection in educational settings is primarily dictated by the unique characteristics of the instructional context and the prevailing conditions. The planning and implementation of lessons with and about AI-supported measurement and data acquisition requires basic knowledge of how AI-supported measurement and data acquisition can be used in the classroom to facilitate learning. This understanding enables students to autonomously collect data themselves and employ intelligent data pre-processing techniques, thereby maximizing their learning potential. However, despite all the possible benefits of incorporating AI in the classroom, in particular the access to knowledge acquisition, a critical examination of its implementation from a technical and socio-cultural standpoint should always be encouraged.

The competence area **Data Acquisition** (AI.DAQ) describes the individual ability to collect data directly or indirectly with digital tools by entering data, digitizing analog data, taking images, and making films, using probes, sensors, and programs (or apps), and obtaining data from documentation media such as images or videos. The use of AI technologies in digital data acquisition is possible, for example, through AI-controlled sensors, automated signal pre-processing, pattern recognition and data extraction from images and

Table 8
Competencies in the area of *Data Acquisition* (AI.DAQ).

Competence level	Teaching (TPACK)	Methods, Digitality (TPK)	Content-specific context (TCK)	Special Tools (TK)
Name	AI.DAQ.T.N1: Name suitable alternatives to scientific AI-based DAQ for use in schools. AI.DAQ.T.N2: Name AI-based DAQ and associated measurement strategies for specific teaching/learning scenarios of different scenarios for appropriate use (appropriate to the addressee, subject and target), e.g., for the automated collection of large amounts of data when. • investigating changes in skin temperature during sport or smoking using thermography with thermal imaging cameras, • determining the nitrate content of a body of water using AI-based data acquisition, • analyzing movements in swarms of insects with mobile devices.	AI.DAQ.M.N1: Name possible further aspects that the use of AI-based DAQ can affect in learning and teaching, e.g., with regard to • expenditure of time, • forms of organization, • forms of presentation, • methods, • media knowledge/familiarization, • personal and social consistency.	AI.DAQ.C.N1: Name scientific scenarios and, if applicable, contexts of AI-based DAQ, e.g., • detection of anomalies in microscopic images, • evaluation of images of microbiological culture plates, • particle collisions in acceleration experiments, • use of "Chemical Robotic Labs". AI.DAQ.C.N2: Name measuring instruments with AI-based DAQ (e.g., automated experimentation) that meet the current requirements of scientific research. AI.DAQ.C.N3: Name the measurement systems corresponding to AI.DAQ.C.N2 and relevant safety standards, e.g., conductivity measurements with AI-supported signal decoding to identify and quantify individual components of a mixture.	AI.DAQ.S.N1: Name possibilities of AI-based DAQ, e.g., • for species identification, • for image recognition, • in the automated aggregation of existing raw data.
Describe (including necessary procedures)	AI.DAQ.T.D1: Describe didactic prerequisites for the use of AI-based DAQ systems in the classroom (e.g., individually adapted instructions), effects of AI-based DAQ on the respective teaching methods, access to basic skills, knowledge acquisition and NOS concepts made possible by AI systems.	AI.DAQ.M.D1: Describe pedagogical requirements as well as advantages and disadvantages that arise methodologically when using AI-based DAQ, e.g., with regard to the aspects listed under AI.DAQ.M.N1.	AI.DAQ.C.D1: Describe selected scientific scenarios of AI-based DAQ using examples, e.g., automated detection of tumor cells in tissue samples.	AI.DAQ.S.D1: Describe at least one technical implementation option for AI-based DAQ applications, including the necessary procedure with reference to current hardware and software and the associated standards. AI.DAQ.S.D2: Describe the measurement characteristics (e.g., measurement range, measurement accuracy, resolution, sampling rate, areas of application, limitations) of AI-based systems.
Use/Apply (practical and functional realization)	AI.DAQ.T.A1: Planning of complete teaching-learning scenarios with the integration of an AI-based DAQ and consideration of suitable social organization of the classroom and modes of (social) interaction. AI.DAQ.T.A2: Implementation of complete teaching-learning scenarios with the integration of an AI-based DAQ and consideration of suitable social organization of the classroom and modes of (social) interaction.		AI.DAQ.C.A1: Carry out the recording of measured values in a scientific context using AI-based DAQ.	AI.DAQ.S.A1: Carry out the measurement acquisition for at least one example of the above-mentioned possibilities of AI-based DAQ.

videos. In each case, digitality-related aspects must also be given greater consideration.

In addition to the above definition of the competency area *Data Acquisition*, Table 8 provides guidance on the competencies expected of teachers in this competency area.

5.7. Data processing (AI.DAP)

The processing of digital data can be supported by AI in a variety of ways. Specifically, AI algorithms have the capacity to process and analyze substantial volumes of data, identify intricate patterns, and develop predictive models. The automation of repetitive data processing tasks is a potential benefit of AI, which can result in enhanced efficiency. AI-based tools can be used to process various types of data types, including image, audio and video. For instance, AI algorithms can analyze and interpret image data, identify objects or faces within the image. Natural language processing (NLP) can be used to analyze texts in order to extract information or recognize moods and opinions. Additionally, spoken language can also be converted into text and translated into different languages. In the context of video data, AI algorithms can facilitate the tracking of movements and the identification of events. However, the use of AI in data processing is also associated with risks, such as systematic distortions due to unrepresentative or incomplete training data or overinterpretation of correlations as causal relationships.

Building on its ability to process and interpret diverse data formats, AI is becoming increasingly central to scientific research, particularly in response to the growing volume and complexity of research data. Advancements in modern measurement methods and the increasing automation of processes are among the primary factors contributing to the proliferation of requiring analysis. The employment of AI is essential for the effective management of large data sets, as these systems possess the capability to expeditiously discern patterns within substantial data sets and systematically categorize data. This enables the prediction of properties for previously unstudied substances and the generation of design proposals for new drugs or materials [e.g., 38, 86,87]. In biochemical research, the quest for inhibitors of various disease-associated proteins can be significantly shortened by leveraging AI as part of an so-called "in silico screening" [25,28,31]. In the field of battery research, AI-based data processing has been employed to develop new types of electrode materials [86].

Given the increasing integration of AI in scientific research, it is equally important to reflect on how these developments translate into educational context. In scenarios involving the implementation of AI-based data processing for teaching and learning purposes, it is important to ensure the transparency of the procedural steps and the relevant laws, regulations, and data security requirements, particularly in contexts involving personal data. In addition, the strengths and weaknesses of various AI algorithms, their IT hardware requirements and their associated risks must be thoroughly addressed. The aim should be to promote the responsible and ethical handling of data among learners and to avoid placing excessive trust in AI systems without human verification. Appropriate use in teaching scenarios should aim to cultivate a fundamental understanding of the principles underlying AI-based data processing in suitable scientific contexts, along with

Table 9
Competencies in the area of *Data Processing* (AI.DAP).

Competence level	Teaching (TPACK)	Methods, Digitality (TPK)	Content-specific context (TCK)	Special tools (TK)
Name	AI.DAP.T.N1: Name tools for the appropriate use (appropriate to the addressee, subject and target) of AI-based DAP. AI.DAP.T.N2: Name scenarios for using the aforementioned possibilities of AI-based DAP in specific teaching-learning situations with a fit to a meaningful context in terms of content.	AI.DAP.M.N1: Name the necessary prior knowledge and competencies of the learners necessary for a teaching-learning scenario in order to use the AI-based techniques. AI.DAP.M.N2: Name methodological aspects of learning and teaching via AI-based DAP, e.g., with regard to • expenditure of time, • form of organization, • equipment and material requirements, • algorithms used, • data security. AI.DAP.M.N3: Name points to consider when processing personal data in the context of work steps (with AI-based technologies).	AI.DAP.C.N1: Name established procedures of AI-based data processing in the field, e.g., the application of • supervised learning, • unsupervised learning. AI.DAP.C.N2: Name scientific scenarios and associated methods of subject-specific AI-based DAP, e.g, • pattern recognition in (large) data sets, • prediction (of binding affinities, protein folding, etc.), • design proposals (e.g., molecule design or material design), …on the basis of own data or from databases.	AI.DAP.S.N1: Name different data types and encodings as well as associated data and file formats (and the operations permitted with them) for an AI-based DAP, e.g., for • image and video, • audio, • values (integer, float), • text. AI.DAP.S.N2: Name various algorithms that are used in AI-based DAP applications. AI.DAP.S.N3: Name AI-based tools for • filtering, • calculation of new variables, • preparation for visualization, • statistical analysis, • image, audio and video analysis, • linking of data, • automation in data processing, • pattern recognition, and • processing of large amounts of data. AI.DAP.S.N4: Name supported file formats of AI-based tools. AI.DAP.S.N5: Name possibilities of AI-based export and import of digital data of the data types and encodings mentioned. AI.DAP.S.N6: Name options for AI-based conversion of data and data formats. AI.DAP.S.N7: Name options for preparing data for AI-based data processing. AI.DAP.S.N8: Name the importance of the quality of the data and the resulting influence of the AI model.
Describe (including necessary procedures)	AI.DAP.T.D1: Describe didactic requirements of AI-based DAP for use in and effects on the respective teaching methods. AI.DAP.T.D2: Describe access to basic competencies made possible by AI-based DAP (especially in the area of knowledge acquisition).	AI.DAP.M.D1: Describe ways to protect and anonymize personal data. AI.DAP.M.D2: Describe the advantages and disadvantages of methodological aspects of AI-based DAP in learning and teaching, e.g., with regard to • expenditure of time, • form of organization, • equipment and material requirements, • algorithms used, • data security.	AI.DAP.C.D1: Describe scientific scenarios with associated methods in which subject-specific AI-based DAP occurs: • pattern recognition (e.g., movement data from insect swarms), • prediction (of binding affinities, protein folding, etc.), • design proposals (e.g., molecule design or material design), … on the basis of own data or from databases.	AI.DAP.S.D1: Describe AI-based methods for • filtering, • calculation of new variables, • preparation for visualization, • statistical analysis, • image, audio and video analysis, • linking of data, • automation in data processing, • pattern recognition, and • processing of large amounts of data. AI.DAP.S.D2: Describe the possibilities of AI-based data processing. AI.DAP.S.D3: Describe the difficulties of AI-based data processing. AI.DAP.S.D4: Describe the advantages and disadvantages of AI-based data processing.
Use/Apply (practical and functional realization)	AI.DAP.T.A1: Planning of full teaching scenarios with the integration of AI-based DAP and the consideration of suitable social organization of the classroom and modes of (social) interaction. AI.DAP.T.A2: Implementation of full teaching scenarios with the integration of AI-based DAP and the consideration of suitable social organization of the classroom and modes of (social) interaction.			AI.DAP.S.A1: Apply AI-based methods for • filtering, • calculation of new variables, • preparation for visualization, • statistical analysis, • image, audio and video analysis, • linking of data, • automation in data processing, • pattern recognition, and • processing of large amounts of data. AI.DAP.S.A2: Export and import digital data of data types and formats with AI-based software.

its capabilities and limitations. This understanding should be facilitated through appropriate and professional learning experiences involving AI-based technology in science lessons. Teachers must exercise caution in the ethical and data protection-compliant management of data and must educate their students to question AI-based results, thereby cultivating the capacity to discern any AI-induced distortions. Particular attention should be placed on the acquisition of fundamental competencies in the individual scientific disciplines through the utilization of AI-based data processing methodologies.

> **The competence area Data Processing (AI.DAP)** describes the individual ability to process data with digital tools. This includes filtering, calculating new variables, processing, statistical analysis and merging data sets. The use of AI technology is particularly beneficial for automation, pattern recognition and efficient processing of large amounts of data. Digitality-related aspects must also be increasingly taken into account.

Table 9 summarizes the competency expectations for teachers in the competency area of data processing.

5.8. Simulation and modeling (AI.SIM)

In addition to theory and experiment, simulation has become a third essential method for generating scientific knowledge. Models enable the analysis of systems and the prediction of how processes develop over time. Such simulations, which are predominantly computer-aided, can attain considerable complexity when a substantial number of parameters must be taken into account, demanding significant computing power. AI, in contrast, enables the analysis and processing of large and complex data sets, thereby enabling the identification of patterns and the generation of forecasts, even in the absence of an analytical model. Increasingly, modern AI systems combine machine learning with simulation, resulting in so-called simulation AI models that can significantly reduce computational demands. Conversely, simulations can support AI development by generating training data, especially in domains where empirical data is scarce. AI algorithms can thus support or enable simulations and modeling in various ways: by accelerating data processing, e.g., pattern recognition in large data sets, optimizing simulation parameters, and improving the efficiency of model calculations.

The creation of simulations and models based on scientific data – retrieved from various databases such as PDB, Reaxys or PubChem – represents a rapidly expanding field of research. One prominent example is the AI-based platform AlphaFold, which predicts three-dimensional protein structures based on amino acid sequences [e.g., 24,26,30]. This technique significantly accelerates biomedical research and enhances accuracy of structural predictions, which is crucial for understanding biological mechanisms and developing new therapies [27]. In materials science, AI is employed to promote the development of novel materials, for example in the additive manufacturing of metals [34]. Similarly, in the context of sustainable chemistry with production of specific chemicals and fuels, AI is used to simulate enzyme level concentrations to optimize biochemical production processes [88]. The field of chemistry, specifically organic chemistry with pharmacological research, benefits from AI-based planning of synthesis routes by means of retrosynthesis [e.g., 20,37]. In the domain of biology, specifically genetics and molecular biology, AI systems contribute to designing promoters for targeted gene expression control [89], optimizing genome sequencing processes [90], and predicting certain protein presences from gene sequence [91]. Physics also sees widespread AI adoption, including for example models of high-energy particle collisions in order to obtain information about particle interaction [36]. Furthermore, AI-based modeling and simulations play a key role in, e.g., climate modeling and meteorology. It helps improve cloud parameterization in climate simulations [92] and enhances precipitation forecasting accuracy [93].

The integration of AI-based simulations and modeling into teaching-learning scenarios opens up new possibility for analyzing large data sets with students and uncovering patterns or trends that would remain hidden, or require significant effort using traditional approaches As a result, even complex simulations or modeling can be conducted within a reasonable time frame without the need for specialized IT infrastructure. Moreover, AI can be used to generate data that would otherwise be unavailable or difficult to obtain in school contexts to teachers and students. When used appropriately, AI-based simulations and modeling offer valuable opportunities for professional and meaningful learning with, through and about AI-based technology in science education. However, this requires teachers to not only select AI-based tools that are suitable in terms of didactics and pedagogy, but also scientifically sound and capable of delivering accurate and valid simulations. In addition, the necessary and well-functioning technical infrastructure must be in place to be ensured that all students have equitable access to these tools. Teachers must also address both data protection and ethical guidelines in both data generation and analysis. Finally, it is essential to reflect with students on the potential social and cultural implications of AI-generated simulations and models to foster critical and responsible engagement with the technology.

> The competence area **Simulation and Modeling** (AI.SIM) describes the individual skills to perform computer-aided modeling and to use existing digital simulation AI models in a targeted and addressee-oriented manner for the knowledge acquisition and communication process as well as the knowledge of limits and potentials of models and simulations in the process of knowledge acquisition. Digitality-related aspects must also be taken into account to a greater extent.

Table 10 summarizes the expectations of science teachers in the competency area of *Simulation and Modeling*.

6. DiKoLAN AI in teaching practice at university and school

The curricular integration of AI into both the subject-specific and interdisciplinary parts of teacher training at universities is proving to be a complex challenge in terms of educational and university didactics. A never-ending "wave" of AI tools is entering the education sector (see e.g., https://theresanaiforthat.com), whether as learning and working tools in the hands of students or as support for teachers in planning and designing lessons. In light of this rapid development, teacher education must adapt accordingly – yet the curricular response remains inconsistent. The skills required for the pedagogically meaningful and subject-specific didactic usage of AI must be systematically developed during the course of study, in line with the roles and responsibilities of both teachers and learners. Currently, the curricular inclusion of AI in teacher training is still variable and often fragmented, with limited standardization between institutions and education systems [94]. Some institutions are beginning to take a more proactive approach by offering targeted training and support for teachers, staff, and students [18]. While comprehensive integration across all disciplines and phases of teacher education curricula across disciplines is still its early stages, pioneering programs are emerging where AI content forms a central component of the curriculum. For example, AI tools are introduced in courses and AI-related tasks are used to prepare prospective teachers for the pedagogically meaningful use of AI in the classroom [95,96].

Existing frameworks for digitalization-related and/or AI-related competence descriptions (see Section 3.3) can be used as a structuring tool for designing modern, comprehensive, and cumulative teacher education - for the field of natural sciences, this is provided by the DiKoLAN AI orientation framework. The DiKoLAN framework concept is increasingly being adopted in the natural science subjects of teacher training. A curriculum development based on DiKoLAN AI opens up a visionary perspective for the future design of teaching-learning processes. Within this vision, the relevance of DiKoLAN AI not only increases for future didactic concepts but can also be applied retrospectively to currently emerging and established teaching concepts for describing competencies. Huwer et al. [71] describe a total of 20 practical examples from various science subjects that illustrate the versatility of DiKoLAN AI. The distribution of these examples reflects the cross-disciplinary and subject-specific integration of AI-related teaching concepts: biology didactics (6 examples), chemistry didactics (5 examples), physics didactics (4 examples), computer science didactics (3 examples) and digital education and media technology (2 examples).

The diversity of the concepts for workshops, training or teaching events in the natural sciences brought together in this single work highlights the relevance of a broad-based competence framework, as shown by the eight competence areas in DiKoLAN AI. A heat map analysis of these practical examples reveals a range of approaches to AI-related competencies, from comprehensive coverage of all DiKoLAN AI areas to a selective focus on promoting specific competencies (see Fig. 4).

6.1. Use of AI as a learning object in higher education to a teaching tool in professional teaching practice

If the practical examples cited in Huwer et al. [71] reflect the current situation of courses offered on AI at universities, it appears that teaching concepts adopting a broad, overarching perspective on the use of AI in education still predominate (for now). Mostly with a pedagogical and significantly less subject-specific focus, the initial aim here is to clarify the state of knowledge regarding the potential of AI in lesson planning and design on the basis of subject-specific contexts and scenarios, and also to reflect on limitations [e.g., 12,97].

Given the wide-ranging applications for AI, a correspondingly large number of different skills are also being developed among prospective teachers. AI is an increasingly prominent subject in teacher training: it is being explored and developed for specific subjects, tested in application, and reflected upon. Not only are AI tools used as examples to help (prospective) teachers practice didactically meaningful use, but also the basics of AI. From the underlying algorithms to practical applications and social and ethical implications [77,98], AI in its full breadth is at the center of the teaching process. In a reflective application, the teaching goal is to enable prospective teachers to use AI as a teaching tool in their future professional field [e.g., 99,100]. The didactic relevance

Table 10
Competencies in the area of *Simulation and Modeling* (AI.SIM).

Competence level	Teaching (TPACK)	Methods, Digitality (TPK)	Content-specific context (TCK)	Special tools (TK)
Name	AI.SIM.T.N1: Name scenarios for the appropriate use of AI-based simulations and modeling as well as software and strategies for use in a specific teaching-learning scenario, e.g., • as a way of gaining knowledge, • in the absence of other affordable, accessible and safe methods, • as a subject-specific working method, • as a time-optimized form of data acquisition, • as an approach for targeted, variable model criticism.	AI.SIM.M.N1: Name advantages and disadvantages, typical characteristics and limitations of AI-based simulation/modeling in teaching/learning scenarios, taking into account, e.g., • technical correctness (simplification), • model variant, normative, descriptive (weather report), • quality of the representation, • time required (calculation time), • training time, • the realization of risk-free, error-tolerant spaces (safety aspects), • the respective mathematical models (e.g., parameters, rounding errors, input accuracy), • necessary prior knowledge AI.SIM.M.N2: Name advantages and disadvantages of AI-based simulation/modeling compared to analog and other digital (non-AI-based) simulations.	AI.SIM.C.N1: Name several content-specific scientific scenarios in which AI-based simulation or modeling is used to gain knowledge (e.g., climate models, gene sequences, protein structure, prediction of material properties, optimization of cell strain designs for the production of chemicals and fuels in microbial cells, design of molecules with desired properties). AI.SIM.C.N2: Name several methods of AI-based simulation or modeling in research scenarios (e.g., particle collisions, protein structures, inverse molecule design). AI.SIM.C.N3: Name several data sources from which data that can be used for AI-based modeling can be obtained (e.g., substance or protein databases, weather data, populations, measured values from the specialist sciences). AI.SIM.C.N4: Name insights gained with AI-based simulations (e.g., protein structures, inverse molecule design, weather forecasting, climate models, global warming, material properties). AI.SIM.C.N5: Name different target categories for the use of AI-based simulations: • predictive → generation of measured values, • analytical → comparison of measured values, • visualization → communication. AI.SIM.C.N6: Name different target categories for the use of AI-based modeling applications: • predictive → generation of measured values, • analytical → comparison of measured values, • visualization → communication.	AI.SIM.S.N1: Name several AI-based technologies that can be used for simulations and modeling. AI.SIM.S.N2: Name the data basis, skills and prior knowledge required by the user for digital modeling, e.g., with regard to • programming and syntax, • required hardware (performance), • data pool size for calculations, • machine learning methods. AI.SIM.S.N3: Name several AI-based simulations and approaches to such simulations: • for generating data in the cognitive process, • for comparing with experimentally obtained data, • for illustrating subject-specific relations.
Describe (including necessary procedures)	AI.SIM.T.D1: Describe didactic prerequisites for the use of AI-based simulations and modeling in the classroom and their effects on the respective teaching methods as well as approaches to basic skills made possible by AI-based systems (especially in the area of knowledge acquisition and, if applicable, communication).	AI.SIM.M.D1: Describe and evaluate AI-based simulations and modeling software in terms of motivation (usability, attractiveness, clarity of description and objectives) and methodology (flexibility, suitability for the target group, implementation, clear description and objectives). AI.SIM.M.D2: Describe advantages and disadvantages of AI-based simulation/modeling compared to analog and other digital (non-AI-based) simulations.	AI.SIM.C.D1: Describe the knowledge gained with AI-based simulations and their advantages/disadvantages as well as their epistemological limitations in various concrete research scenarios.	AI.SIM.S.D1: Describe the functional scope of the technologies mentioned (cf. AI.SIM.S.N1), e.g., with regard to the simulation of patterns in large and/or complex data sets.
Use/Apply (practical and functional realization)	AI.SIM.T.A1: Planning of complete teaching scenarios with the integration of AI-based simulations or modeling and the consideration of suitable social organization of the classroom and modes of (social) interaction. AI.SIM.T.A2: Implementation of complete teaching scenarios involving AI-based simulations or modeling and the consideration of suitable social organization of the classroom and modes of (social) interaction.			

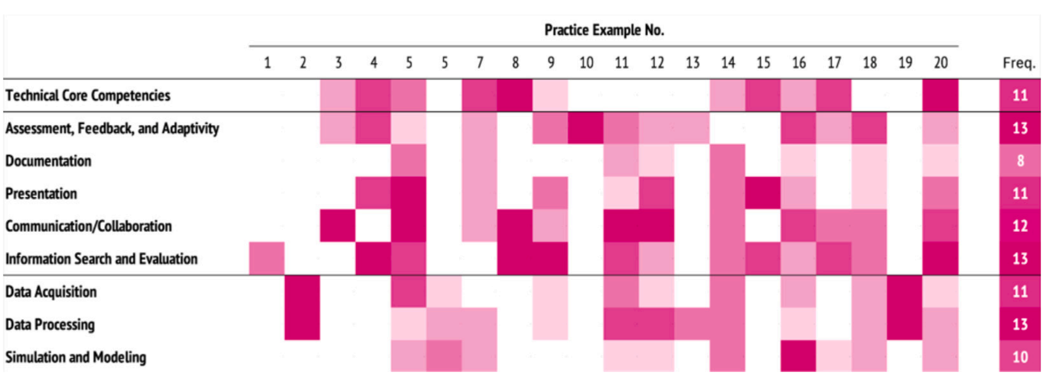

Fig. 4. Heat map analysis to address DiKoLAN AI-related competencies in these practical examples.

of this broad integration of AI lies in recognizing the opportunities and limitations of AI and integrating it responsibly into the respective subject teaching and/or applying it in the preparation and follow-up of subject teaching. In this sense, teachers use AI applications primarily to create teaching materials efficiently – for example, presentations, texts or videos. Recent studies show that AI is also used in lesson planning, in assessing of student performance, and in the creation of feedback [101].

In light of the dynamic development in the field of AI and its growing relevance for educational contexts, it is therefore essential that teacher education not only imparts technological competencies, but also systematically integrates a reflective engagement with the pedagogical, ethical, and subject-specific implications of AI.

6.2. Use of AI as a teaching and learning tool in higher education to specifically promote skills

Teaching concepts in which AI functions as a tool to support or enable teaching by lecturers or to support the learning of student teachers are still rare in higher education. In the heatmap analysis of practical examples (Fig. 4) from Huwer et al. [71], those concepts stand out due to their selective addressing and the more in-depth development of selected DiKoLAN AI competencies. In addition to AI chatbots as a sounding board for teaching experiments [102] or AI tools in the design of a project [103], AI is also used in a much more subject-specific way, in which AI is used specifically to develop and promote subject-related learning content. Examples are photosynthesis, plant identification, or the acquisition of scientific knowledge [104–106] that is used by prospective teachers or as part of acquiring competence in such usage. Overall, implementation scenarios in which AI is deployed within subject-specific domains of competencies in the natural sciences, such as digital data acquisition, data processing and simulation/modeling, remain scarce, even in the increasingly developed field of subject-didactic teaching. Therefore, the differentiated and operationalized competencies in DiKoLAN AI can provide orientation and guidance in the development of subject-specific didactic teaching concepts in order to implement both broad and targeted competence development. The practical use of DiKoLAN AI in higher education is immanent – already evident from the wide range of practical examples presented in Huwer et al. [71] – and creates a sustainable basis for preparing future teachers to effectively integrate AI into science teaching.

7. Limitations

The DiKoLAN AI framework was developed through a multi-stage, iterative process involving experts from the DACH region and multiple cycles of feedback (alpha-, beta-, and gamma-testing). While this approach ensured that the framework was grounded in expert opinion and informed by diverse disciplinary perspectives, several limitations should be acknowledged. First, the development process only incorporated contributions from 9 experts (+ symposium audience) during the alpha-testing, 24 experts during the beta-testing and 55 experts during the gamma-testing. Feedback was obtained through workshops, discussions, and written comments/questionnaires. As a result, the framework reflects the informed perspectives of experienced stakeholders in the DACH region.

Second, the case studies referenced in this manuscript were not designed or conducted to empirically validate the DiKoLAN AI framework. Instead, they are based on contributions from multiple authors who presented their own teaching activities on AI in science education. These activities were subsequently mapped to the competence areas of DiKoLAN AI to illustrate that the framework fulfills its intended purpose—namely, to describe the competences relevant for science teachers and to provide a structure for categorizing and characterizing such teaching activities. This mapping demonstrates the framework's potential to enable, for the first time, systematic comparison of AI-related teaching in science education and to support alignment across different courses. The case studies therefore serve solely to demonstrate the applicability and descriptive capacity of DiKoLAN AI, not to offer empirical evidence of its effectiveness.

Third, the focus of the framework is currently limited to science education within the DACH region (Germany, Austria, Switzerland), and its applicability to other subjects or international contexts remains to be tested. Cultural, institutional, and curricular differences can require adaptation of specific areas of competence.

Finally, the framework assumes a certain level of digital infrastructure and institutional readiness that may not be present in all teacher training environments. Its implementation will therefore depend on contextual factors such as available resources, teacher expertise, and support structures for professional development.

8. Conclusion

The increasing integration of AI in research, society, and education is not only a technological challenge, but also a social and didactic one. In the natural sciences in particular, AI offers new opportunities for knowledge acquisition, experimentation, and adaptive learning support. The rapidly increasing importance of AI in scientific research is also bringing new contexts into the classroom, namely learning not only with AI, but also about AI.

Given the rapid technological development and the increasing presence of AI technologies in education, a structured approach to developing teachers' subject-specific skills and formulating corresponding job-related AI skills is of central importance.

Therefore, the development of the DiKoLAN AI framework is a valuable step toward the structured integration of AI in science teacher training. DiKoLAN AI draws on existing models such as TPACK and DigCompEdu and aims to extend them by incorporating AI-specific competencies relevant to science education. A distinguishing feature of DiKoLAN AI is that, unlike general frameworks such as DigCompEdu, it is subject-specific, AI-focused, and deliberately integrates three interdependent dimensions: science didactic competencies for designing meaningful AI-supported learning processes, technological competencies for effectively using and understanding AI tools, and socio-cultural competencies for critically reflecting on AI's societal implications. This triad constitutes a key strength of the framework and ensures a holistic preparation of science teachers for the pedagogical, technical, and ethical challenges of AI integration. Of particular interest is the inclusion of socio-cultural knowledge dimensions, which complement the predominantly technology-focused perspectives of earlier models and align with the DPACK framework, as well as the emphasis on computer science literacy. This reflects the growing recognition that AI represents not only a technological evolution but also a broader societal shift.

Within DiKoLAN AI, AI-related competencies are tentatively structured along eight core fields of action: evaluation, feedback, and adaptivity; documentation; presentation; communication and collaboration; information search and evaluation; data acquisition; data processing; and simulation and modeling. This framework is intended to offer a structured basis for subject-specific reflection on the integration of AI in science education. Although its design is based on established theoretical models and expert discussions, its practical value and applicability have yet to be empirically validated.

Illustrative case studies from the DACH region have been included to demonstrate the applicability of the framework. These were not designed or conducted to empirically validate DiKoLAN AI. Instead, they consist of teaching activities on AI in science education, presented by different authors, which were subsequently mapped to the competence areas of DiKoLAN AI. This mapping shows that the framework achieves its intended purpose: to describe the competencies relevant for science teachers and to serve as a tool for categorizing and characterizing such teaching activities. By doing so, it enables, for the first time, the comparison of AI-related teaching in science education and supports alignment across different courses. The case studies therefore serve solely to illustrate the descriptive and structuring capabilities of DiKoLAN AI, rather than to provide evidence of its effectiveness.

For the future of science teaching, it is crucial that teachers not only have a basic understanding of AI technologies but are also able to integrate them into their subject lessons in a didactically meaningful way. One of the strengths of DiKoLAN AI is its flexibility. It can support curriculum development, help with teacher self-assessment, and provide a structured framework for designing professional development programs. This versatility makes it valuable not only for educators but also for policymakers and institutions that aim to systematically integrate AI competencies into science education. The practical and scientifically based DiKoLAN AI framework offers a structured and subject-specific approach that can contribute to curriculum development in teacher training and serve as a reference framework for further training programs and the evaluation of AI-related educational processes. The exemplary case studies from the DACH countries demonstrate the applicability and relevance of the framework in specific teaching and learning scenarios. Future research should investigate how well the framework can be transferred to other countries, what country-specific adaptations may be necessary, and whether it can be applied internationally, as seen with DiKoLAN. Additionally, it will be important to develop concrete assessment tools similar to those used in DiKoLAN to measure teachers' competency levels, allowing for targeted post-qualification support. For a sustainable and systematic integration of AI-related skills into the education system, it will be crucial to disseminate this framework throughout teacher training programs. It must also be continuously refined to align with emerging technological and social developments and evaluated through empirical research. This approach is essential to adequately prepare both teachers and students for the challenges and opportunities of an increasingly AI-driven society.

CRediT authorship contribution statement

Johannes Huwer: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Christoph Thyssen:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sebastian Becker-Genschow:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lena von Kotzebue:** Writing – review & editing, Writing – original draft, Investigation. **Alexander Finger:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Erik Kremser:** Writing – review & editing, Writing – original draft, Formal analysis. **Sandra Berber:** Writing – original draft, Formal analysis. **Mathea Brückner:** Writing – original draft, Formal analysis. **Nikolai Maurer:** Writing – original draft, Formal analysis, Conceptualization. **Till Bruckermann:** Writing – review & editing, Writing – original draft, Conceptualization. **Monique Meier:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Lars-Jochen Thoms:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT (https://chat.openai.com; GPT-4o) as well as DeepL (https://www.deepl.com) and Grammarly (https://www.grammarly.com) in order to improve the readability and language of single sentences. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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